



Freie Universität Berlin
Department of Physics

Quantum speed limit dependence on the number of local controls in a qubit array

A thesis presented for the degree:

Master of Science

by

David Anthony Pohl

First Reviewer: Prof. Dr. Christiane Koch

Second Reviewer: Prof. Dr. Dirk Manske

Submitted on August 17, 2024

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Acknowledgments

I would like to express my gratitude to Prof. Dr. Christiane Koch for giving me the opportunity to write this thesis and continue working on my Bachelor's project. I am continually impressed by her profound grasp of the subject matter. In addition, I would like to express my gratitude to Matthias Krauss and Fernando Gago-Encinas for their guidance and support in developing my programming skills and theoretical knowledge, which proved invaluable in enabling me to produce and reflect on my results. I am also indebted to Dr. Daniel Basilewitsch and Dr. Steve Campbell for their patience in answering my questions and stimulating discussions. Finally, I would like to thank my colleagues Yimeng Wang and Karl Horn for their companionship and camaraderie.

Abstract

Universal quantum computing requires operator controllability of the qubit array. This is typically realized via qubit-qubit couplings and local external controls on every qubit which becomes challenging when scaling to large numbers of qubits. It was recently shown that the number of external controls can be reduced to the extreme limit of a single control [1]. However, this comes at the expense of longer gate durations. Here, we investigate the gate duration depending on the number of local controls. In particular, we show that reducing controls increases the quantum speed limit (the shortest time to generate a quantum gate). We determine this limit for a universal set of gates for different 3-qubit systems using quantum optimal control.

Chapter 1

Introduction

Quantum computers build on the principles of quantum mechanics, utilizing quantum bits or qubits. Unlike classical computers that use bits limited to 0 or 1, quantum computers employ qubits, which can exist in multiple states simultaneously due to quantum superposition. This allows quantum computers to process extensive information in parallel, potentially leading to faster solutions for specific problems. Another unique property exhibited by qubits is quantum entanglement. When qubits become entangled, the state of one qubit is directly related to the state of another, regardless of the distance between them. This enables quantum computers to perform complex operations more efficiently and potentially share information faster than classical systems. Due to their potential to solve specific problems much more efficiently than classical computers, quantum computers are considered to revolutionize computations. Many quantum algorithms provide an exponential speed up compared to classical algorithms. For instance, consider the computational task of finding prime factors of integer numbers. While the most efficient known classical algorithm runs in exponential time, the quantum analog Shor's algorithm solves the same problem in polynomial time [2]. This could be revolutionary for the field of cryptography, since algorithms that secure information on classical computers are based on the paradigm that prime factorization of large integer numbers needs exponential runtime, that Shor's algorithm could efficiently break. This has led to increased interest in developing quantum-resistant cryptographic techniques [3]. The field of quantum computing is rapidly changing, employing a variety of physical systems as platforms for quantum information processing [4–8]. There are some general challenges associated with the physical implementation of quantum computers. For example, they are highly susceptible to errors due to environmental noise and imperfect qubit control. The former is characterised by unwanted interactions between the environment and the qubits. The delicate nature of quantum states makes them susceptible to decoherence, where quantum information is lost due to interactions with the environment, leading to computational errors. In the worst case, noise-induced decoherence can lead to the complete loss of quantum information, destroying the quantum advantage. In addition, the development of robust error correction techniques is essential to maintain the precision and reliability of quantum computation [9–11]. Imperfect qubit control, often attributed to low pulse accuracy, leads

to gate errors, which introduce inaccuracies in the final quantum state. Consequently, increasing the pulse accuracy is imperative as it effectively mitigates the gate errors. Several works have focused on the implementation of low-error unitary gates through the application of optimal control techniques. [12–16]. The complexity of controlling individual qubits is increasing as efforts are made to build large-scale quantum computers. Superconducting quantum circuits have emerged as a candidate for scalable quantum information processing in terms of physical implementation [8]. In 2019, for example, Google unveiled the Sycamore quantum processor. This architecture consists of 53 superconducting transmon qubits, tunably coupled to 4 neighbours [17]. To elaborate on the quantum processor, it allows for universality, which means that any desired unitary gate can be implemented, see chapter 2.3. In terms of implementation speed, certain one- and two-qubit gates can be executed in tens of nanoseconds. In particular, the fastest gate can be executed within a short duration of 12 ns [17]. In terms of control mechanisms, each qubit is locally controlled by a microwave pulse and a magnetic flux that tunes the central qubit frequency. In addition, there is a practical property that we would like to call the array limit, i.e. the maximum number of qubits that the architecture can support. This number cannot be increased due to technical constraints. It is important to note that system universality, implementation speed and the array limit are linked to the configuration of the system’s local controls. Regarding its problem-solving capabilities, Google claimed that Sycamore could solve a particular computational task faster than any classical machine on the planet ¹. Although Sycamore is regarded as a significant milestone in the field of quantum computing, the computational task it has solved is not yet practically relevant. Furthermore, it faces several unsolved challenges that are typical for quantum processors of its generation, which ultimately impede its widespread practical application. With regard to scalability, for instance, there are cutting-edge processors that have recently achieved a scale surpassing 130 qubits (IBM Heron). However, the problem of scalability persists[18]. To elaborate further, this challenge extends beyond mere physical wiring; it involves intricate signal routing at the quantum chip level, constraints related to space and cooling power within cryogenic systems, and the introduction of additional noise channels. Collectively, these factors contribute to an exacerbation of the scalability issue [17, 19].

The difficulty of scaling the qubits could be considerably reduced by the identification of optimised qubit array architectures that facilitate scalability. A promising approach to maintaining universality could be to simplify the architecture, for example by reducing the number of microwave wires while ensuring that superconducting qubits still interact with neighbouring qubits in a specific manner. Building upon this concept, I conducted research for my Bachelor’s thesis to ascertain whether qubit arrays with less local controls than qubits could still achieve universality. A notable discovery was made, indicating that it is feasible to strategically design a universal 3-qubit array with just a single local control, provided that the interactions between the qubits can be selected in a desired manner. Moreover, an extension

¹Nevertheless, this outcome has been questioned on the grounds that IBM, which was responsible for the operation of the supercomputer at the time, suggested that it could potentially resolve the identical issue in a more time-efficient manner than Google had indicated.

of this research by *F. Gago-Encinas et al.* demonstrated the existence of a universal 5-qubit array in the limit of a single local control [1]. The question of the relationship between the minimal number of local controls and the quantum speed limit for universal quantum computing was left open for further investigation. In this context, the quantum speed limit refers to the minimum time required to execute a quantum gate. Moreover, it was postulated that a reduction in the number of local controls would result in longer gate durations, thereby bringing the system closer to the decoherence limit. Furthermore, it was argued that the time required to execute a gate would be equal to or longer than the time required with additional controls added to the system. Conversely, reducing the number of local controls must be balanced with the necessity of sufficiently fast logical operations. It can thus be postulated that a trade-off exists between implementation speed and the number of local controls. This thesis aims to provide insight into the relationship between implementation speed and the number of local controls in qubit arrays. In particular, we explore the quantum speed limit for elements of a universal set in dependence on local controls in certain qubit arrays. The following questions provide a framework for our enquiry:

- Can it be confirmed whether a larger number of controls results in a shorter quantum speed limit ?
- Are there gates that are naturally faster than others ?
- Which gates determines the quantum speed limit of that universal set ?
- Can we identify inherently fast control positions within a qubit array ?

The responses to these questions could prove instrumental in the advancement of high-performance, scalable quantum computers where the trade-off between scalability and implementation speed is optimally balanced. Furthermore, an investigation into the quantum speed limit of unitary gates is of considerable importance in optimising the performance of quantum algorithms.

Chapter 2

Universal Quantum Computing

2.1 The DiVincenzo Criteria

This section is based on the framework developed by D. DiVincenzo in 2000 [20]. The DiVincenzo criteria articulate five fundamental requirements that a physical system must manifest to be eligible as a universal quantum computer. The criteria represent idealizations, their realization presents challenges for viable platforms for quantum computing.

1. A scalable physical system with well-characterized qubits.

To enable quantum computing, it is necessary to have a physical system that incorporates a collection of qubits. A qubit, in essence, represents a quantum two-level system, for instance, the two spin states of an electron or the vertical and horizontal polarization of a single photon. Most quantum systems differ from these "natural qubits"¹, since they have much more than two levels. However, multi-level quantum systems can effectively be used as qubits, by confining all dynamics to two quantum levels, usually the ground and the first excited state of the system. Conventional notation for both qubit states are $|0\rangle$ and $|1\rangle$. A generic single-qubit state is an element of a two-dimensional complex Hilbert space and can be written as

$$\alpha |0\rangle + \beta |1\rangle, \quad \alpha, \beta \in \mathbb{C}, \quad (2.1)$$

satisfying the normalization constraint $|\alpha|^2 + |\beta|^2 = 1$ [21]. Thus, the distinguishing characteristic of a qubit, setting it apart from a classical bit, lies in the superposition principle. This principle enables the existence of an infinite range of potential qubit states.

A generic two-qubit state is represented by a 4-dimensional vector:

$$\alpha |00\rangle + \beta |01\rangle + \gamma |10\rangle + \delta |11\rangle, \quad \alpha, \beta, \gamma, \delta \in \mathbb{C} \quad (2.2)$$

for which holds $|\alpha|^2 + |\beta|^2 + |\gamma|^2 + |\delta|^2 = 1$. A two-qubit state is defined as an entangled state if it cannot be expressed as a tensor product of two individual qubit states. This implies

¹A natural qubit has exclusively two quantum levels provided by nature.

that an entangled state can only be represented as a superposition in any local basis. When a qubit-qubit interaction is present (e.g. $\hat{H}_{int} \sim \hat{\sigma}_x^{(1)} \hat{\sigma}_x^{(2)}$), the state of the two qubits evolves naturally to an entangled state, which indicates that the qubits are entangled.

Considering n qubits, the general state is specified by a 2^n -dimensional complex vector. Note, that the Hilbert space dimension grows exponentially with the number of qubits.

The designation of a qubit as "well-characterized" means various aspects: Primarily, it implies precise knowledge of all physical parameters, including the internal Hamiltonian dictating the energy eigenstates of the qubit. This understanding extends to recognizing the presence and couplings of other states, interactions with neighboring qubits.

2. The ability to initialize qubit states to a simple fiducial state.

In classical computing, the registers of a central processing unit must be set to a known value before executing computations. Similarly, quantum registers, comprising a collection of qubits, must be prepared in a known state before executing a quantum protocol. This preparatory step is referred to as the initialization of the qubit states. Furthermore, initialization is mandatory for quantum error correction. Which requires a continuous, fresh supply of qubits in a low-entropy state. There are two main approaches to setting qubits to a standard state: the system can either be cooled when the ground state of its Hamiltonian is the state of interest, or the fiducial state can be achieved by a measurement which projects the system onto that. Qubit initialization stands as an important task in the operation of quantum devices, currently serving as a bottleneck for device clock speed across numerous quantum hardware platforms. Several pertinent studies have addressed the challenge of optimizing qubit initialization [15, 22, 23].

3. Long relevant decoherence times

Decoherence plays a crucial role in the foundational principles of quantum physics, serving as the primary mechanism facilitating the transition to classical behavior [24]. In addition, decoherence presents a significant challenge in the field of quantum computing. The advantage of quantum computing over classical computing relies on utilizing resources such as entanglement and superposition. However, these quantum resources can only flourish under conditions of sufficiently coherent dynamics. Decoherence arises from the interaction between qubits and their surrounding environment. If decoherence persists for extended periods, it can severely diminish the capability of the quantum computer, ultimately rendering it comparable to a classical machine: Consider the single qubit state from equation (2.1), written as density matrix,

$$\rho_0 = |\psi\rangle \langle\psi| = |\alpha|^2 |0\rangle \langle 0| + \alpha\bar{\beta} |0\rangle \langle 1| + \bar{\alpha}\beta |1\rangle \langle 0| + |\beta|^2 |1\rangle \langle 1|. \quad (2.3)$$

The off-diagonal entries of a pure state quantify coherence. Nevertheless, once the decoherence time has passed, these off-diagonal entries vanish. Consequently, the state shifts from a quantum superposition to a classical superposition,

$$\rho_0 \mapsto \rho(T_{\text{dec}}) = |\alpha|^2 |0\rangle \langle 0| + |\beta|^2 |1\rangle \langle 1|. \quad (2.4)$$

Furthermore, if coherence vanishes within larger qubit arrays, quantum entanglement becomes unattainable.

It has been emphasized that relevant decoherence times should be sufficiently long. In the context of quantum particles, various degrees of freedom can possess distinct decoherence times. However, not all of these times are equally significant depending on the circumstances. For instance, while the electron's position state may experience rapid decoherence in a solid-state environment, its spin coherence time could be considerably longer. In certain scenarios, it can be arranged that only one particular decoherence time is relevant for quantum computation, such as the spin coherence time. The determination of the relevant decoherence time is influenced by the choice of qubit basis states, such that certain degrees of freedom, like orbital states, may be deemed irrelevant for certain computations. Maintaining quantum coherence over time is crucial for the accuracy of quantum computations. Qubits in a quantum computer should ideally exhibit long relevant decoherence times, indicating stability and the ability to preserve quantum information for extended periods. It has been repeatedly emphasised by leading researchers in the field that decoherence times must be considerably longer than the "clock time" of a quantum computer, which represents the time required to execute a single quantum gate. This is essential to facilitate the practical and reliable implementation of quantum computation. For example, Preskill addresses the thresholds for reliable quantum computation, which are implicitly related to the ratio of decoherence time to clock time [25]. A widely accepted guideline is that the ratio of decoherence time to gate time should be greater than or equal to one. If this ratio is greater than or equal to 10^3 , then there is sufficient time to perform a substantial number of gate operations, and error correction strategies can effectively mitigate errors.

4. A universal set of quantum gates.

A universal quantum computer can be programmed using a universal set of gates². This set should enable the generation of any quantum operation, offering the flexibility necessary for a wide range of quantum algorithms. A quantum algorithm is defined as a sequence of unitary matrices, each acting on a small number of qubits:

$$\prod_{i=1}^N \hat{U}_i = \hat{U}_N \hat{U}_{N-1} \dots \hat{U}_2 \hat{U}_1. \quad (2.5)$$

Implementing the algorithm requires initially identifying Hamiltonians capable of producing the necessary unitary transformations. To elaborate, one needs to identify the set of Hamiltonians $\{\hat{H}_i\}$ that generate the unitary matrices required for the quantum algorithm $\{\hat{U}_i \simeq e^{-\frac{i}{\hbar} \hat{H}_i t}\}$. Executing a unitary operation is typically achieved using external classical

²It is important to note that gate-based quantum computing is not the only avenue for achieving universal quantum computing. There are a number of potential approaches. One alternative is to perform operations on continuous variables, such as the position and momentum of light modes. This approach, known as continuous-variable quantum computing, can be universal when equipped with the appropriate set of operations, including both Gaussian and non-Gaussian transformations.

control fields (pulses) on the qubits. Running a quantum algorithm requires a "pulse sequence" designed to activate $\hat{H}_1$ from time 0 to time t , $\hat{H}_2$ from time t to time $2t$, and so on. However, the ability to switch Hamiltonians on and off is not trivial at all. It heavily depends on the physical implementation of qubits and the available control mechanisms. This requirement is met when the qubit register can be sufficiently controlled. Thus, it is imperative to conduct a controllability analysis, see chapter 2.3. Further, determining the pulses that execute the gates is also significant and will be discussed in chapter 3.4 and chapter 3.3.

5. A qubit-specific measurement capability.

The result of a quantum computation must be read out, requiring the ability to measure specific qubits. Consider a qubit density matrix:

$$\rho = |\alpha|^2 |0\rangle \langle 0| + \alpha \bar{\beta} |0\rangle \langle 1| + \bar{\alpha} \beta |1\rangle \langle 0| + |\beta|^2 |1\rangle \langle 1| \quad (2.6)$$

$$= p_0 |0\rangle \langle 0| + \alpha \bar{\beta} |0\rangle \langle 1| + \bar{\alpha} \beta |1\rangle \langle 0| + (1 - p_0) |1\rangle \langle 1|. \quad (2.7)$$

An ideal measurement of the qubit should give outcome $m = 0$ with probability p_0 and $m = 1$ with probability $p_1 = 1 - p_0$ independent of any other parameters of the system, including the state of nearby qubits, and without changing the state of the rest of the quantum computer. Such an ideal measurement is said to have 100% quantum efficiency, while in contrary real measurements always have less. While the fidelity of a quantum measurement is not captured by a single number, the single quantum-efficiency parameter is often a very useful way to summarize it. Similarly, the decoherence time is useful to (partially) summarize the damage caused to a quantum state by the environment. While quantum efficiency of 100% is desirable, much less is needed for quantum computation, since there is a trade off possible between quantum efficiency and other resources which results in reliable computation.

A key characteristic of a quantum computer is its clock speed, which dictates how quickly a quantum processor can complete tasks. The time required to execute a task, such as a quantum gate or algorithm, is influenced by several factors, including qubit initialization time, gate duration, measurement and readout time, and qubit reset time. This thesis focuses on the quantum speed limit of gates, as reducing gate duration can improve quantum algorithm performance. Ideally, if each gate operates at its theoretical speed limit, it would boost the overall clock speed of the quantum computer.

2.2 Gate-Based Quantum Computing

This section begins with a basic example, the free evolution of a qubit, but presents a different viewpoint: treating the free evolution of a qubit as a sequence of gates acting on an initial state. We will discuss the principles of gate implementation and explore key examples for single and two-qubit gates. Additionally, we will explain the motivation behind the Hamiltonian modeling of a simplified system of N superconducting qubits.

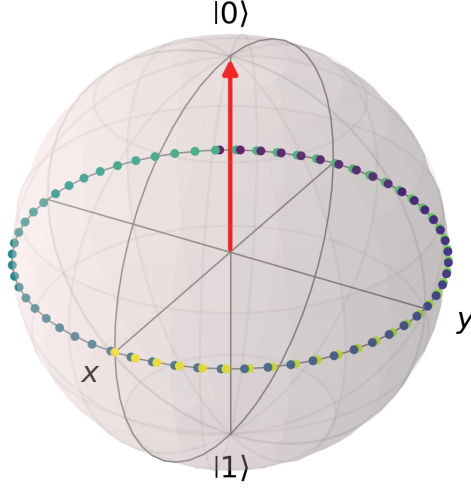


Figure 2.1: The free evolution of the qubit state $|\psi(t_0)\rangle$, as governed by the Hamiltonian in Eq. (2.8), is illustrated. This evolution shows the qubit state rotating around the z-axis of the Bloch sphere. For this plot, the parameters used are $\omega = 1$ and $T = 10$.

$$\hat{H} = -\frac{\omega}{2}\hat{\sigma}_z \quad (2.8)$$

and prepared in the state $|\psi(t_0)\rangle = |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. The evolution generated by the Hamiltonian in Eq. (2.8) is represented by

$$\hat{U}(t) = e^{i\frac{\omega}{2}\hat{\sigma}_z t} \equiv \begin{pmatrix} 1 & 0 \\ 0 & e^{-i\omega t} \end{pmatrix}. \quad (2.9)$$

Hence, as time elapses, the state $|\psi(t_0)\rangle$ rotates around the z-axis, in the equatorial plane of the Bloch sphere see Fig. 2.1. The evolution operator $\hat{U}(t)$ varies in the group $U(2)$, since Eq. (2.9) represents a unitary matrix at all times t .

$$\hat{U}(t) |\psi(t_0)\rangle = \hat{U}(t_N) \dots \hat{U}(t_2) \hat{U}(t_1) \hat{U}(t_0) |\psi(t_0)\rangle. \quad (2.10)$$

That is why each matrix $\hat{U}(t_i)$ in Eq. (2.10) represents an infinitesimal rotation that acts on the qubit state at time t_i . In gate-based quantum computing, a unitary matrix that acts on a qubit state vector is called a unitary gate. Moreover, the evolution operator reaches a class of unitary gates in (2.9), that are called phase gates. This rather simple example illustrates that the free evolution of a qubit, can be considered as a (continuous) gate sequence acting on the initial qubit state.

The discussion below covers a range of single qubit gates widely employed in quantum information theory [26]: The Pauli gates X, Y, Z , associated with the Pauli matrices $\hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z$, are oftentimes employed for unitary gates that operate on a single qubit.

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (2.11)$$

The Hadamard gate and the T gate are additional examples of single-qubit gates,

$$H = \frac{1}{\sqrt{2}}(X + Z) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{4}} \end{pmatrix}. \quad (2.12)$$

Among the discussed gates, the free qubit evolution in Eq. (2.9) reaches the T gate at $t = \frac{\pi}{4\omega}$ and Z gate at $t = \frac{\pi}{\omega}$. Nevertheless, gates, such as X, Y and H , are unreachable, unless the system is altered by a suitable control. Adding controls provides the possibility of changing the evolution and hence to implement certain gates.

$$H(t) = -\frac{\omega}{2}\hat{\sigma}_z + u(t)\hat{\sigma}_x. \quad (2.13)$$

Through skillful design of the control pulse $u(t)$ and reasonable choice of the control operator, any desired unitary gate can be implemented, as we will discuss in section 2.3. Moreover, a description of how gates are implemented using optimal control is given in chapter 3.4.

In gate-based quantum computing, quantum algorithms are constructed as sequences of unitary gates as defined in Eq. (2.5). Nevertheless, the free evolution and the controlled evolution of a single qubit are conceptionally different, because a quantum algorithm is applied to a collection of N qubits, or qubit register. Therefore, all gates can be represented by elements of the special unitary group $SU(2^N)$ or the unitary group $U(2^N)$ ³. Furthermore, a certain quantum algorithm requires a specific set of unitary gates, that operate in a strict sequential order, generally with different gate durations, designed to solve a computational problem [17].

The gate-based approach has been successfully implemented on superconducting qubit registers, which motivate our model system. The Hamiltonian in Eq. (2.14) models a qubit register with time-independent qubit-qubit couplings and local controls manipulating each qubit state individually, which is the subject

$$\hat{H}(t) = \underbrace{\sum_{i=1}^N -\frac{\omega_i}{2}\hat{\sigma}_z^{(i)}}_{\text{drift}} + \underbrace{\sum_{i \neq j} \sum_{k,l}^3 \kappa_{(k,l)}^{i,j} \hat{\sigma}_k^{(i)} \hat{\sigma}_l^{(j)}}_{\text{qubit-qubit interactions}} + \underbrace{\sum_{i=1}^3 \sum_{m=1}^3 u_{(m,i)}(t) \hat{\sigma}_m^{(i)}}_{\text{controls}} \quad (2.14)$$

In this model, the first two terms constitute the time-independent physical system to be manipulated by the controls, which denotes the last part. The drift characterizes the level-spacing of all N qubits, as it represents a scenario without qubit-qubit interactions and switched-off controls, ω_i describes the central frequency of the i 'th qubit.

Furthermore, the qubit-qubit interactions denote a non-zero coupling between two qubits. A constant single qubit-qubit interaction is mediated via the operator $\sigma_k^{(i)} \sigma_l^{(j)}$ and its prefactor

³The physical difference both groups relates to the fact $U(2^N)$ contains elements of the form $U = e^{i\phi} \mathbb{1}_{2^N}$, $e^{i\phi} \in \mathbb{C} \setminus \mathbb{R}$. Those are elements that are proportional to the identity matrix which exhibit a non-real global phase. However, in a quantum system, the global phase is physically irrelevant. Hence, both representations of universal gates are equally satisfactory.

$\kappa_{(k,l)}^{i,j}$, representing the strength of the qubit-qubit interaction.

In a manner akin to Eq.(2.13), the local controls are acting on single qubits via local operators of the form $\sigma_m^{(i)}$, where $u_{m,i}(t)$ represents a control field. Utilizing this model, one can simulate the implementation of unitary gates on an qubit register. Thereby, the constant qubit-qubit interactions generate entanglement, which presents both challenges and opportunities. It can create challenges for the control strategy, when the entanglement is unwanted and could provide opportunities if the figure of merit is an entangling operation. Due to their ability to use quantum resources such as entanglement, quantum algorithms may offer computational benefits in selected tasks. These advantages stem from the employment of entangling gates. Here, the CNOT gate and the B gate are provided as examples for two-qubit gates, which are gates specifically designed to entangle pairs of qubits.

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad (2.15)$$

$$B = \begin{pmatrix} \cos(\frac{\pi}{8}) & 0 & 0 & i \sin(\frac{\pi}{8}) \\ 0 & \cos(\frac{3\pi}{8}) & i \sin(\frac{3\pi}{8}) & 0 \\ 0 & i \sin(\frac{3\pi}{8}) & \cos(\frac{3\pi}{8}) & 0 \\ i \sin(\frac{\pi}{8}) & 0 & 0 & \cos(\frac{\pi}{8}) \end{pmatrix}. \quad (2.16)$$

The CNOT gate is frequently encountered in the field of quantum information, due to its relatively simple structure and efficiency. For instance, it can be proven that any arbitrary two-qubit gate can be approximated using only 3 applications of the CNOT gate [27]. However, the lesser-known B gate surpasses the CNOT gate in terms of gate count and experimental feasibility: Just two applications of the B gate suffice to approximate any arbitrary two-qubit gate. Moreover, in setups utilizing superconducting qubits, the B gate can be implemented faster than the CNOT gate [27].

Current state-of-the-art quantum processors feature a variety of qubit array structures, which differ in size, interactions, geometry, and control configurations. Building a universal qubit array is challenging, as universality depends on the number of qubits, the nature of qubit-qubit interactions, and the available controls. The next section, 2.3, will focus on evaluating whether a qubit array possesses this universal property.

2.3 Controllability and Universality

In quantum computing, universality relates to an N -qubit array's ability to generate all unitary gates. Hence, all $U \in U(2^N)$ or all $U \in SU(2^N)$ can be executed. Remarkably, this feature is unique to certain qubit arrays. To ascertain whether a qubit array possesses the property of universality, it is necessary to determine whether all (special) unitary matrices are included in the reachable set of the given qubit array. Equivalently, one can study if

the corresponding generators of all (special) unitary matrices form a complete basis of the operator space. The matrix A is called skew-Hermitian, if it fulfills $A = -A^\dagger$. Additionally, a skew-Hermitian matrix A is called a generator of the unitary matrix U , if $e^A = U$. Consider for instance, the unitary matrix in Eq. (2.16). Moreover, it can be expressed in exponential form in terms of its generator G .

$$B = e^G, \quad G = i\frac{\pi}{4}\hat{\sigma}_x^{(1)}\hat{\sigma}_x^{(2)} + i\frac{\pi}{8}\hat{\sigma}_y^{(1)}\hat{\sigma}_y^{(2)} \quad (2.17)$$

In principle, the generator of a unitary matrix is not unique, and there are instances where a unitary matrix possesses multiple generators. Therefore, the B gate can be represented through a variety of generators. For example, $\tilde{G} = \frac{i\pi}{8}\hat{\sigma}_y^{(1)}\hat{\sigma}_x^{(2)} + i\frac{3\pi}{8}\hat{\sigma}_y^{(1)}\hat{\sigma}_y^{(2)}$.

This section is principally based on by the third chapter of [28]. To streamline controllability analysis, unitary matrices are regarded as a Lie group, while their corresponding skew-Hermitian generators form a Lie algebra. A Lie group possesses both the algebraic structure of a group and the geometric structure of a smooth manifold [28–30]. The group operations of inversion and multiplication are analytic functions, implying that they can be approximated by a Taylor expansion at every point within their range and every neighborhood. The unitary group $U(N)$ is a Lie group, and the special unitary group $SU(N)$ is a Lie subgroup of $U(N)$, as it constitutes a subgroup algebraically and a submanifold geometrically. On a broader scale, a Lie group is linked to a so-called Lie algebra in a manner akin to the connection between a unitary matrix and its generator: Lie algebra elements are generators of Lie group elements. Here, a Lie algebra $\mathcal{L}$ can be regarded as a matrix vector space over the field $\mathbb{R}$, maintaining closure under operations involving the commutator $[\cdot, \cdot]$. For $A, B, C \in \mathcal{L}$ and $\alpha \in \mathbb{R}$, the following conditions must hold:

1. Bilinearity

$$\begin{aligned} [A + B, C] &= [A, C] + [B, C]. \\ [A, B + C] &= [A, B] + [A, C]. \\ \alpha[A, B] &= [\alpha A, B] = [A, \alpha B]. \end{aligned}$$

2. Skew-Symmetry

$$[A, B] = -[B, A].$$

3. Jacobi-Identity

$$[A, [B, C]] + [B, [C, A]] + [C, [A, B]] = 0.$$

The Lie algebra $\mathfrak{u}(N)$ is associated with the Lie group $U(N)$. Formally, $\mathfrak{u}(N)$ is the $\mathbb{R}$ -vector space of skew-Hermitian matrices equipped with the commutator:

$$\mathfrak{u}(N) := \{A \in \mathbb{C}^{N \times N} | A = -A^\dagger\}. \quad (2.18)$$

Furthermore, $\mathfrak{u}(N)$ has an important Lie subalgebra, which is the Lie algebra of traceless-skew hermitian matrices,

$$\mathfrak{su}(N) := \{A \in \mathfrak{u}(N) | \text{Tr}(A) = 0\}. \quad (2.19)$$

The connection between these Lie groups and their Lie algebra is given through the matrix exponential [29]. Let $A \in \mathbb{C}^{N \times N}$, then

$$\mathbf{e}^A = \sum_{k=0}^{\infty} \frac{A^k}{k!} = \mathbb{1} + A + \frac{A^2}{2} + \dots \quad (2.20)$$

The matrix exponential maps (traceless) skew-Hermitian matrices into (special) unitary matrices. Therefore, it is customary to express the Lie group in terms of the Lie algebra using the matrix exponential, leading to the following representation (Lie group–Lie algebra correspondence) [28]:

$$\mathbf{e}^{\mathfrak{u}(N)} = U(N), \quad \mathbf{e}^{\mathfrak{su}(N)} = SU(N). \quad (2.21)$$

Consider a quantum system with a Hamiltonian of the form

$$\hat{H}(t) = \hat{H}_0 + \sum_{k=1}^M \hat{H}_k u_k(t) \quad (2.22)$$

Here, $\hat{H}_0$ denotes the quantum system, that is subject to M controls $u_k(t)$ mediated by the operators $\hat{H}_k$. A special case of such a system is the N-qubit array in Eq. (2.14). One can define the Lie algebra associated with this quantum system containing information its possible dynamics:

$$\text{Lie}\{i\hat{H}(t)\} = \text{Lie}\{iH_0, iH_1, \dots, iH_M\}. \quad (2.23)$$

The dynamical Lie algebra is constructed as the span of all linearly independent nested commutators. Starting with all Hamiltonians $\{iH_0, iH_1, \dots, iH_k\}$ in Eq. (2.22), a nested commutator takes the form $L = [K_1, [K_2, \dots [K_p, K_{p+1}] \dots]]$. Each L is assigned a depth p , indicating the number of required commutators to obtain L . Beyond a certain depth, no new linearly independent element can be constructed, implying that only linearly dependent elements can be obtained. The dynamical Lie algebra $\text{Lie}\{H(t)\}$, is a Lie subalgebra of $\mathfrak{u}(N)$. The algorithm to generate a basis of a dynamical Lie algebra is outlined as follows [28]:

1. Step 0: Consider all linear independent matrices in $i\hat{H}(t)$ as a linear span from the set $\{i\hat{H}_0, i\hat{H}_1, \dots, i\hat{H}_M\}$ (commutators of depth 0)
2. Step $n \geq 1$:
 - (a) Calculate the commutator with input matrices from nested commutators of depth $n - 1$ and depth 0 (obtained at step $n - 1$ and step 0).
 - (b) Consider the commutators up to depth n and remove all linearly dependent nested commutators.
 - (c) Stop if N^2 or $N^2 - 1$ linear independent nested commutators have been obtained.

A quantum system represented by Eq. (2.22) undergoes unitary evolution, wherein the Schrödinger operator equation takes the form of a bilinear equation.

$$\dot{\hat{U}} = -i\left(\hat{H}_0 + \sum_{k=1}^M \hat{H}_k u_k(t)\right)\hat{U}(t), \quad \hat{U}(0) = \mathbb{1}_N. \quad (2.24)$$

With a particular choice of the controls $u_k(t)$, the solution to Eq. (2.24) results in a sequence of unitary matrices for all times t . This sequence constitutes the reachable set $\mathcal{R} = \{\hat{U}(t)\}_{0 \leq t \leq T}$. Furthermore, this reachable set $\mathcal{R}$ has the structure of a Lie group, which is a Lie subgroup of $U(N)$. Remarkably, the Lie group $\mathcal{R}$ is connected to the dynamical Lie algebra through the matrix exponential. Analogously to Eq. (2.21), one can express

$$\text{Lie}\{i\hat{H}(t)\} \mapsto \mathcal{R} = e^{\text{Lie}\{i\hat{H}(t)\}} \quad (2.25)$$

The Hamiltonian dictates the accessibility of certain Lie algebra basis elements, that are computed according to the previous presented algorithm, that span the dynamical Lie algebra. Further, the Hamiltonian also dictates the unitary evolution and thereby the reachable set of unitary matrices that form the connected Lie group. Equation (2.25) uncovers that both structures are also linked through an Lie group–Lie algebra correspondence. Thus, Eq. (2.25) illustrates how the reachability of unitary matrices in the Lie group relates to the accessibility of Lie algebra basis elements. Hence, access to all generators requires a full dynamical Lie algebra and is aligned with a full Lie group of the quantum system, which motivates the following theorem:

Theorem 2.3.1 (*Lie Algebra Rank Condition and Operator Controllability*) *Let Eq. (2.24) be the dynamical equation associated to a quantum system. Let $\mathcal{L}$ be the dynamical Lie algebra and $\mathcal{R} = e^{\mathcal{L}}$ its Lie group of reachable unitary matrices. The quantum system is said to be **operator controllable**, if one of the following two conditions can be satisfied.*

1. $\dim(\mathcal{L}) = \dim(\mathfrak{u}(N)) = N^2$
2. $\dim(\mathcal{L}) = \dim(\mathfrak{su}(N)) = N^2 - 1$

Additionally, it is said that the Lie algebra achieves full rank.

To emphasize, a full rank of the Lie algebra denotes that the quantum system is operator controllable. Implying all N^2 entries of the evolution operator are simultaneously controllable, which implies the ability to arbitrarily steer the dynamics of this quantum system [28, 31]. Operator controllability carries significant implications, as it implies the existence of optimal controls capable of generating any desired unitary evolution. However, it is important to interpret the last sentence with caution, as operator controllability typically assumes access to sufficient physical resources and imposes no time constraints. This implies that operator-controllable systems cannot be arbitrarily manipulated within short times [32], which we will cover in chapter 3.4. In the case of an N-qubit array being the quantum system under

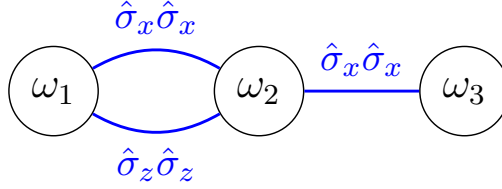


Figure 2.2: The asymmetric chain is a 3-qubit array described by the Hamiltonian in Eq. (2.26) and resembles a universal qubit array. For more details regarding testing universalitay see 2.3

investigation, then Eq. (2.25), illustrates how the availability of generators in the Lie algebra relates to the reachability of unitary matrices and thus to the possibility to execute unitary gates on the N-qubit array. Therefore, the calculation of the dynamical Lie algebra establishes a universality test: In the case of a full Lie algebra rank, all generators are accessible, all unitary matrices are reachable and thus all unitary gates are executable. Furthermore, universality implies that any gate $\hat{V} \in SU(2^N)$ can be approximated via a finite sequence $\hat{U}_k \dots \hat{U}_2 \hat{U}_1$ to arbitrary precision with $\hat{U}_i$ from a finite set $\{\hat{U}_i\}$ called universal set. It should be noted that universal sets are not unique and can be selected in such a way that they consist solely of single- and two-qubit gates, with no additional multi-qubit gates. This is related to the fact, that any multi-qubit gate can be represented as a sequence of single-qubit and two-qubit gates [33]. For instance, H , T and B constitute a universal set [12], while H , T and CNOT represent another example [26]. If the physical implementation allows the availability of multiple universal sets, it may be advisable to select one that is particularly efficient. For aforementioned reasons the B gate can be considered to be more efficient than the CNOT gate. Additionally, H and T are considered efficient, because one can approximate any single-qubit gate to arbitrary precision ϵ with an relative adequate efficiency of $O[\log^4(\frac{1}{\epsilon})]$ with a sequence that only consists of H and T [34].

The following example illustrates a operator controllable qubit array, that will also be focus of analysis in chapter 4.2. We utilize a special case of Eq. (2.14) to model a three-qubit system that only couples through nearest-neighbour interactions, which results in a line-shaped geometry, see Fig. 2.2. The Hamiltonian of the asymmetric chain reads,

$$\hat{H}_A = \sum_{i=1}^3 -\frac{\omega_i}{2} \hat{\sigma}_z^{(i)} + \kappa_{(x,x)}^{1,2} \hat{\sigma}_x^{(1)} \hat{\sigma}_x^{(2)} + \kappa_{(z,z)}^{1,2} \hat{\sigma}_z^{(1)} \hat{\sigma}_z^{(2)} + \kappa_{(x,x)}^{2,3} \hat{\sigma}_x^{(2)} \hat{\sigma}_x^{(3)}. \quad (2.26)$$

Moreover, the model allows for the selection of distinct interaction strengths, thereby enabling the consideration of non-uniform couplings. Additionally, the choice of different coupling strengths is a realistic approach, given that in a real chain of qubits, the effective interaction strength experienced by neighbouring qubits may also vary due to differences in physical distance between them.

The system features two coupling terms on the left-hand side and a single coupling term on the right-hand side. The unequal distribution of coupling terms may appear to be an inconvenient feature, but it is motivated by mathematical considerations. The couplings

are selected in order to ensure operator controllability in a variety of configurations of $\hat{\sigma}_x$ -controls, which differ in quantity and position. Moreover, the minimum number of coupling terms required for operator controllability in this context is three. There are seven potential configurations for the local controls, denoted by $\{u(t)\hat{\sigma}_x^{(i)}\}$. These can be classified into three cases. The first case, the single control configuration, refers to a setup where only one qubit is individually controlled, with the other qubits remaining uncontrolled. In regard to the system described by Eq. (2.26), the configuration that permits the control of a single qubit is as follows:

$$\mathbf{CP1} : u(t)\hat{\sigma}_x^{(1)}. \quad (2.27)$$

$$\mathbf{CP2} : u(t)\hat{\sigma}_x^{(2)}. \quad (2.28)$$

$$\mathbf{CP3} : u(t)\hat{\sigma}_x^{(3)}. \quad (2.29)$$

The abbreviation "CP" stands for "control position", which specifies the locally controlled qubits. Second, in the *double control configuration*, two qubits are locally controlled, which means that there is a single qubit that remains uncontrolled.

$$\mathbf{CP12} : u_1(t)\hat{\sigma}_x^{(1)} + u_2(t)\hat{\sigma}_x^{(2)}. \quad (2.30)$$

$$\mathbf{CP13} : u_1(t)\hat{\sigma}_x^{(1)} + u_3(t)\hat{\sigma}_x^{(3)}. \quad (2.31)$$

$$\mathbf{CP23} : u_2(t)\hat{\sigma}_x^{(2)} + u_3(t)\hat{\sigma}_x^{(3)}. \quad (2.32)$$

Third, the *triple control configuration* is defined by three qubits that are locally controlled:

$$\mathbf{CP123} : u_1(t)\hat{\sigma}_x^{(1)} + u_2(t)\hat{\sigma}_x^{(2)} + u_3(t)\hat{\sigma}_x^{(3)}. \quad (2.33)$$

Moreover, since the system is operator controllable it allows for the implementation of a universal set of quantum gates, such as

$$\mathcal{U} = \{H^{(1)}, H^{(2)}, H^{(3)}, \quad T^{(1)}, T^{(2)}, T^{(3)}, \quad B^{(1,2)}, B^{(1,3)}, B^{(2,3)}\}. \quad (2.34)$$

To conclude, regardless of the control configuration, it is possible to implement any gate $\hat{V} \in SU(8)$ on the asymmetric chain. For instance, by constructing sequences of the universal gates in Eq. (2.34) to approximate $\hat{V}$.

The speed at which gates can be implemented is expected to vary for different control configurations. It can be argued that the time required to execute any gate will be at least as long as the time needed to perform the same operation with additional controls. This is because more controls can access dynamics that fewer controls cannot reach within the same timeframe. Conversely, reducing the number of local controls generally increases the time required for gate execution [1]. Table 2.1 illustrates the highest depth of the dynamical Lie algebra for the asymmetric chain under various control configurations. We hypothesize that controllability analysis could help determine if a specific control configuration enables faster gate implementation. Typically, controllability analysis provides the depth d of a generator, indicating the number of nested commutators needed to determine this generator. Some

	CP1	CP2	CP3	CP12	CP13	CP23	CP123
max. depth	10	12	16	10	8	8	6

Table 2.1: Maximal depth of the dynamical Lie algebra of the asymmetric chain for different control configurations.

studies suggest a proportional relationship between the depth of a generator and the time required to execute the associated gates [35].

However, it could be proposed that analyzing the maximum depth obtained in a controllability analysis may be sufficient to assess gate implementation speed. Furthermore, we observe the trend that the dynamical Lie algebra of the asymmetric chain has a shorter maximum depth when more controls are utilized and a longer maximum depth when less controls are utilized. To illustrate, the asymmetric chain in the CP3 configuration possesses a higher maximum depth than that seen in the CP12 configuration. This may indicate that specific generators are available with greater rapidity within the CP12 configuration. It may be the case that certain unitary gates are executed more rapidly than in the CP3 configuration. It may therefore be posited that the duration of a specific gate is shorter in the CP12 configuration than in the CP3 configuration. However, it is not evident a priori which generators are required for arbitrary gates. Moreover, the generators obtained at depth d can vary considerably between different control configurations. Consequently, the maximum depth of the Lie algebra may not be a reliable indicator. In conclusion, while controllability analysis helps formulate hypotheses about gate execution times, quantum optimal control techniques are significant for estimating the minimum gate duration for various configurations. The shortest time in which a gate can be executed is known as its quantum speed limit. The next chapter will discuss significant research findings on the quantum speed limit and its implications for implementing fast quantum gates.

Chapter 3

Quantum Speed Limit

The section 3.1 presents essential research on the theory of geometrical quantum speed limits and provides an example of how to derive a speed limit for unitary gates. The subsequent section 3.2 motivates a connection between the quantum speed limit and optimal control theory [36]. Furthermore, it is elaborated on an example how the quantum speed limit can be reliably estimated using Krotov’s method. The latter will be explained in its main features in section 3.3, before this chapter closes with the section 3.4, which concerns the implementation of gates at the speed limit.

3.1 Geometric Speed Limit Equations

The historical contextualisation of the quantum speed limit is largely based on the review paper [37]. Furthermore, the derivations of the speed limit equations can be found in [35]. The quantum speed limit can be understood as the shortest possible time required for a quantum state or a quantum gate to be transformed to another transition to another a quantum state or a quantum gate. However, providing a precise and widely accepted definition for the quantum speed limit is challenging due to varying interpretations among authors.

A significant amount of analytical research has been dedicated to the estimation of the quantum speed limit for transitions between quantum states. In this context, the term ‘quantum speed limit’ is used to describe a system property. To elaborate further, it is concerned with an intrinsic timescale for natural evolution, as discussed in the literature [38–40]. The first formulation of a quantum speed limit equation was presented by Mandelstam and Tamm in 1945 [38]

$$\tau \geq \tau_{MT} = \frac{\pi}{2} \frac{\hbar}{\Delta \hat{H}_0}_{|\psi_0\rangle}. \quad (3.1)$$

Assuming $\hat{H}_0$ to be time-independent, than Eq. (3.1) relates to the shortest time, for a state $|\psi_0\rangle$ to evolve into an orthogonal state $|\psi_\perp\rangle$. However, there are cases where the Mandelstamm-Tamm bound produces unreliable results. Uffink showed, that $\Delta \hat{H}_0_{|\psi_0\rangle}$ diverges, even for finite average energy [41].

Margolus and Levitin provided another bound, where they essentially showed that the speed of quantum evolution cannot surpass average energy (above the ground state energy). Likewise, the derivation assumes a constant Hamiltonian, as well as a transition between two orthogonal states [39].

$$\tau \geq \tau_{ML} := \frac{\pi}{2} \frac{\hbar}{\langle \hat{H}_0 - E_0 \rangle_{|\psi_0\rangle}} \quad (3.2)$$

However, the discovery of (3.2) created a paradoxical situation where there appear to be two seemingly independent bounds based on two different physical properties of the same quantum state. Nevertheless, elementary proofs illustrated that both bounds hold true [41]. Consequently, it was simply assumed, without much justification, that the minimal time a quantum system needs to evolve between two orthogonal states is given by

$$\tau_{QSL} := \max\left\{ \frac{\pi}{2} \frac{\hbar}{\Delta \hat{H}_{|\psi\rangle}}, \frac{\pi}{2} \frac{\hbar}{\langle \hat{H} - E_0 \rangle_{|\psi\rangle}} \right\} \quad (3.3)$$

Levitin and Toffoli found that (3.2) and (3.1) are not inherently different but represent two facets of the same concept. Moreover, in 2009, they were able to prove that a system described by $\hat{H}_0$ evolves at the quantum speed limit—the highest possible speed—whenever the unified bound is saturated [40]. Therefore, Eq. (3.3) sets a fundamental limit for unitary transformations between orthogonal states. Saturation of the unified bound means that both bounds yield the same result, which occurs when the standard deviation and the average energy above the ground state energy are equal. For instance, consider the dynamics generated by the Hamiltonian in Eq. (2.10) with the initial state $|\psi(0)\rangle = |+\rangle$, then the state of the system at time $t = \frac{\pi}{\omega}$ is described by the state $|\psi(t)\rangle = |-\rangle$. Furthermore, if we evaluate the unified bound in Eq.(3.3) for Eq. (2.10) and the initial state, it yields $\tau_{QSL} = \max\{\frac{\pi}{\omega}, \frac{\pi}{\omega}\} = \frac{\pi}{\omega}$. Consequently, the natural evolution of this qubit happens at the theoretical allowed minimum, in other words the qubit evolves at the quantum speed limit. Levitin and Toffoli also proved that the unified bound is saturated if the initial state is an equally weighted superposition of the ground state and the highest excited state.

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|E_0\rangle + |E_n\rangle). \quad (3.4)$$

For a state that is not an equal weighted superposition, both bounds will yield a different result. Then the speed limit will be given by the maximum out of both bounds and the system is not evolving at the theoretical minimum. Hence, Eq. (3.3) sets the ultimate speed limit for the evolution between orthogonal states and for time-independent Hamiltonians [37]. There are generalizations of the unified bound, that do not restrict the initial state and final states to be orthogonal, by utilizing

$$d_{FS}(|\psi_0\rangle, |\psi_\tau\rangle) := \arccos(|\langle \psi_0 | \psi_\tau \rangle|). \quad (3.5)$$

The Fubini Study metric can be considered a natural a measure of distance for two quantum states $|\psi_0\rangle$ and $|\psi_\tau\rangle$ in projective Hilbert space. Similar to a sphere, the projective Hilbert

space is a *curved space*, and the geodesic path (shortest distance between two surface points) relates to the Fubini study metric. Utilizing, Eq. (3.5) one can reformulate the speed limit equation in Eq. (3.3), such that it suits state tranformations with arbitrary angles:

$$\tau_{QSL} := \max\left\{\frac{\hbar}{\Delta H_0} d_{FS}(|\psi_0\rangle, |\psi_\tau\rangle), \frac{2\hbar}{\langle H_0 \rangle} d_{FS}^2(|\psi_0\rangle, |\psi_\tau\rangle)\right\}. \quad (3.6)$$

So far, it was assumed that the Hamiltonian is time-independent. Nevertheless, it is also possible to define a speed limit equation that accounts for the explicit time dependence. However, this requires profound comprehension of the projective Hilbert space geometry. In 1993, Anandan and Aharonov provided a framework to understand the evolution of quantum states in a geometric manner [42]. They realized that the set of states a system takes during its evolution, $\{|\psi(t)\rangle\}_{0 \leq t \leq T}$, can be interpreted as a curve $\mathcal{C}$. Furthermore, they determined the length of $\mathcal{C}$:

$$\text{length}(\mathcal{C}) = \frac{1}{\hbar} \int_0^T \Delta \hat{H}(t) dt, \quad \Delta \hat{H}(t) = \sqrt{\langle H(t)^2 \rangle - \langle H(t) \rangle^2}. \quad (3.7)$$

Hence, the geometric distance traveled by a quantum state is related to the integral of the energy uncertainty over the evolution time. To be more precise, they could prove that, the length of the geodesic path is always smaller or equal to the actual path that the system takes. This is known as the Anandan-Aharonov relation [35, 42]:

$$d_{FS}(|\psi(0)\rangle, |\psi(T)\rangle) \leq \frac{1}{\hbar} \int_0^T \Delta \hat{H}(t) dt. \quad (3.8)$$

Then, one utilizes the mean value of the function $\Delta \hat{H}(t)$, denoted by $\overline{\Delta H}$, to derive a speed limit equation,

$$T \geq \frac{\hbar}{\overline{\Delta H}} d_{FS}(|\psi(0)\rangle, |\psi(T)\rangle). \quad (3.9)$$

Note the similarity to the Mandelstamm-Tamm bound in Eq. (3.1), which can be recovered from Eq. (3.9) by considering orthogonal states and a time-independent Hamiltonian. All the considered quantum speed limit equations have in common, that they associate the speed limit time with a distance in projective Hilbert space. That is why the speed limit equations are said to be *geometrical speed limits*.

In this thesis, it is the objective to determine quantum speed limits of unitary gates. Theoretical considerations regarding a geometric speed limit for unitary gates are the subject of modern research. Nevertheless, one example regarding the formulation of a geometric speed limit for unitary gates is going to be elaborated. This derivation is analog to the derivation of Eq. (3.9) and is based on P. Poggi's research [35]. First, one defines a proper metric that relates to the geodesic path for two unitaries U and V in the Lie group manifold $SU(N)$. For example, one could utilize the Fubini-Study metric for unitaries:

$$D_{FS}(U, V) = 2 \arccos\left(\frac{1}{d} |\text{tr}(U^\dagger V)|\right). \quad (3.10)$$

Consider the evolution of a quantum system, generated by $\hat{H}(t)$, from initial time $t = 0$ to final time $t = T$. Further, one utilizes Eq. (3.10), to determine the geodesic path that connects $\hat{U}(0) = \mathbb{1}$ and $\hat{U}(T)$:

$$D_{FS}(\mathbb{1}, \hat{U}(T)) = 2 \arccos\left(\frac{1}{d} |\text{tr}(\hat{U}(T)^\dagger)|\right). \quad (3.11)$$

Subsequently, the Anandan-Aharonov relation for unitary gates is employed, which reads

$$D_{FS}(T) \equiv D_{FS}(\mathbb{1}, \hat{U}(T)) \leq \frac{2}{\sqrt{d}} \int_0^T \|\hat{H}(t)\| dt. \quad (3.12)$$

After the estimation of the integrand $\|\hat{H}(T)\| \leq \overline{\|\hat{H}\|}$, one is able to derive a quantum speed limit equation for unitary gates:

$$T_{QSL} \geq \frac{\sqrt{d}}{2} \frac{D_{FS}(T)}{\overline{\|\hat{H}\|}}. \quad (3.13)$$

Note that, Eq. (3.13) is also geometric speed limit equation, since it depends on $\hat{U}(T)$ only via its distance to the identity matrix. Nevertheless, a meaningful application of the Eqs. (3.9) and (3.13) is contingent upon the accurate estimation of the integrand. Furthermore, the structure of the control Hamiltonian is inconsequential for the computation of T_{QSL} . Conversely, the structure of the control Hamiltonian is fundamental in the optimization of unitary gates. It is therefore beneficial to employ optimal control theory in order to estimate the quantum speed limit.

3.2 Optimal Control at the Speed Limit

In 2009, T. Caneva et al. demonstrated that specific quantum systems could be steered in adherence to the quantum speed limit [36]. To elaborate, a control protocol was designed to steer a quantum system's evolution in the shortest possible time, in accordance with the quantum speed limit constraints. It was demonstrated that the analytically calculated quantum speed limit is in accordance with the numerical OCT result. To elaborate further, the researchers considered a state transformation in a Landau-Zener model, which is described by the Landau-Zener Hamiltonian.

$$\hat{H}_{LZ}(t) = \omega \hat{\sigma}_x + \Gamma(t) \hat{\sigma}_z. \quad (3.14)$$

The detuning $\Gamma(t)$ fulfills the boundary condition of taking a certain value at initial and final time. For instance, $\Gamma(0) = -\Gamma(T) = \frac{n}{\omega}$ with $n = -5, -10, -20$. At initial time the driven two-level system described by Eq.(3.14) is prepared in the instantaneous ground state of $\hat{H}_{LZ}(\Gamma(0))$. Furthermore, the task was to reach the (nearly orthogonal) ground state of $\hat{H}_{LZ}(\Gamma(T))$, denoted by $|\psi_G\rangle$, in the shortest possible duration.

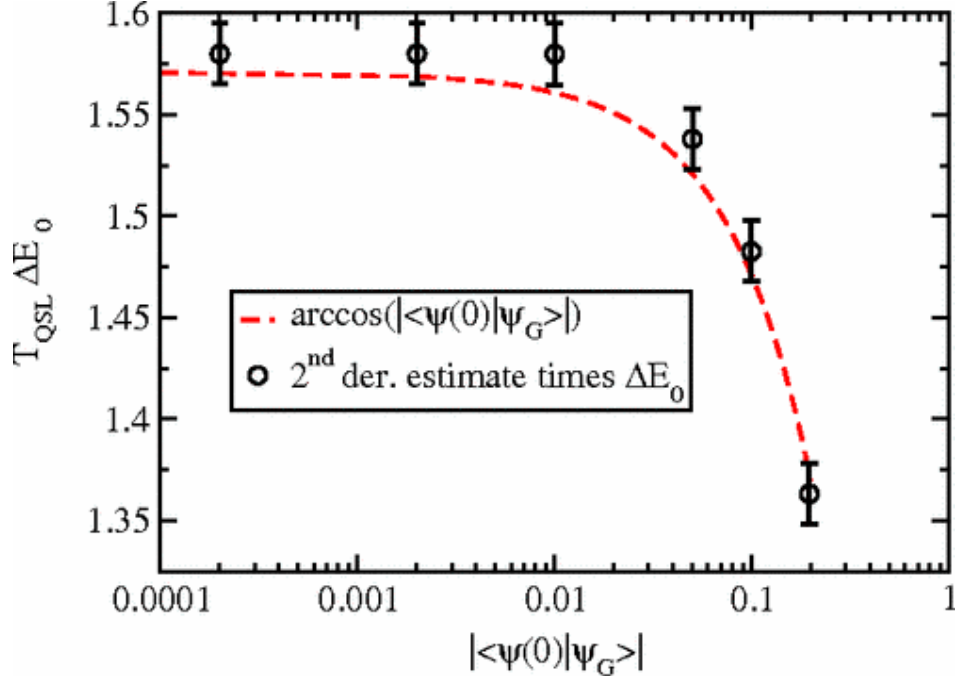


Figure 3.1: This plot compares the OCT-estimated T , determined using the second derivative of the infidelity, with the time-independent estimate T_{QSL} across different $\frac{\Gamma(0)}{\omega}$ ratios. The strong agreement indicates that the efficiency of optimal control is ultimately constrained by the quantum speed limit.

Due to the boundary condition it is possible to neglect the explicit time dependence of the system in the calculation of bound in Eq. (3.6). Consequently, it was found that

$$T_{QSL} = \frac{\arccos(|\langle \psi(0) | \psi(T) \rangle|)}{\Delta E_0} \approx 1.568. \quad (3.15)$$

Moreover, Krotov's method was employed to determine the optimal control fields, that minimize the state infidelity

$$\mathcal{I} = 1 - |\langle \psi(T) | \psi_G \rangle|^2. \quad (3.16)$$

After multiple optimisation processes, one is able to examine the state infidelity as a function of T , see Fig. 3.1. This analysis revealed that the state infidelity $\mathcal{I}$ goes arbitrarily close to zero, whenever $T \geq T_{QSL}$. However, for $T < T_{QSL}$, the infidelity does not converge to zero. In conclusion, it was demonstrated that the application of Krotov's method in a minimal time approach recovers the quantum speed limit in Eq. (3.16). Fundamentally, it reveals a connection between the quantum speed limit and optimal control theory, as it was demonstrated that certain quantum speed limits can be attained using techniques of optimal control theory [36]. This has prompted researchers to employ OCT in contexts where a reliable analytical speed limit equation is unavailable. For instance, a reliable speed limit equation that connects a unitary operation, a quantum system, a set of controls and a speed

limit time is currently not admissible. Despite this, Krotov's method can be employed to estimate the quantum speed limit of unitary gates [13, 43].

3.3 Krotov's Method of Optimization

Krotov's method is an iterative, monotonically convergent optimization algorithm using gradient information to achieve convergence [44]. Furthermore, it can be applied to a variety of optimization problems in quantum physics. Krotov's method can be utilized, for example, to optimize state-to-state transitions in molecular systems [36, 45], to implement gates in superconducting qubit arrays [12, 16, 46] or to optimize towards a target with certain properties in diverse physical settings [47, 48]. The advantage of Krotov's method in comparison to other gradient-based optimization algorithms is that it ensures monotonic convergence [49]. Further, it minimizes a functional $J[\{|\phi_k^{(i)}(t)\rangle\}, \{\epsilon_l^{(i)}(t)\}]$ of the most general form [49], which depends on a set of initial states $\{|\phi_k^{(i)}(t)\rangle\}$ and control fields $\{\epsilon_l^{(i)}(t)\}$ of the i 'th iteration. Numerically, it minimizes a functional that depends on both final -and intermediate times [50]. Typically, the optimization functional consists of three main parts [51],

$$J[\{|\phi_k^{(i)}(t)\rangle\}, \{\epsilon_l^{(i)}(t)\}] = J_T(\{|\phi_k^{(i)}(T)\rangle\}) + \sum_l \int_0^T g_a(\epsilon_l^{(i)}(t)) dt + \int_0^T g_b(\{|\phi_k^{(i)}(t)\rangle\}) dt. \quad (3.17)$$

Here, the first part denotes a final time functional J_T and must be considered a significant part, since it expresses the optimization objective. The second, and third term describe intermediate time functionals. More precisely, g_a expresses field-dependent costs and g_b expresses state-dependent costs. Including state-dependent running cost g_b is usually done when control targets change over time [52] or one wants to penalize specific states [53]. Including field-dependent costs is necessary to encode desired features in the optimized pulses [50]. Now, the fundamental principles underlying the derivation of the update equation are outlined below. The initial step is to present the running cost, which will subsequently be employed in the derivation of the update equation.

$$\begin{aligned} g_a(\epsilon_l^{(i)}(t)) &= \sum_l \frac{\lambda_{a,l}}{S_l(t)} \left(\epsilon_l^{(i)}(t) - \epsilon_{l,\text{ref}}^{(i)}(t) \right)^2 ; \quad \epsilon_{l,\text{ref}}^{(i)}(t) = \epsilon_l^{(i-1)}(t) \\ &= \sum_l \frac{\lambda_{a,l}}{S_l(t)} \left(\Delta \epsilon_l^{(i)}(t) \right)^2. \end{aligned} \quad (3.18)$$

Here, $\lambda_{a,l}$ refers to the inverse step width and $S_l(t) \in [0, 1]$ is the time-dependent shape function. The rationale behind the nomenclature of the parameters will become apparent following the present derivation. Furthermore, it can be proven under the assumption of continuous time, that Krotov's method ensures monotonic convergence [44, 49]. Monotonic convergence means for the functional in Eq. (3.17), that

$$J[\{|\phi_k^{(i)}(t)\rangle\}, \{\epsilon_l^{(i)}(t)\}] \leq J[\{|\phi_k^{(i-1)}(t)\rangle\}, \{\epsilon_l^{(i-1)}(t)\}]. \quad (3.19)$$

This implies that, after a sufficient amount of iterations, the functional approaches a local minimum. A complete derivation for the monotonic convergence, can be found in [50] and in the original work [44]. However, a great didactical discussion is also found in [51]. The condition for monotonic convergence is expressed in terms of the derivative of the state-dependent cost with respect to the control fields [49].

$$\frac{dg}{d\epsilon_l(t)}|_i = 2\Im \left[\sum_{k=1}^N \langle \chi_k^{i-1}(t) | \frac{dH}{d\epsilon_l(t)} |_i | \phi_k^{i-1}(t) \rangle \right]. \quad (3.20)$$

The expression $\frac{dH}{d\epsilon_l(t)}|_i$ relates to the gradient. Selecting the state-dependent cost from Eq. (3.18) yields the update equation, which contains gradient information:

$$\Delta \epsilon_l^i(t)|_i = \frac{S_l(t)}{\lambda_{a,l}} \Im \left[\sum_{k=1}^N \langle \chi_k^{i-1}(t) | \frac{dH}{d\epsilon_l(t)} |_i | \phi_k^{i-1}(t) \rangle \right]. \quad (3.21)$$

The optimized field in iteration i is the result of applying a control update, when starting from the initial guess $\epsilon_l^0(t)$. The update equation reads

$$\epsilon_l^{(i)}(t) = \epsilon_l^{(i-1)}(t) + \Delta \epsilon_l^{(i)}(t). \quad (3.22)$$

Further, Eq. (3.21) implies

$$\Delta \epsilon_l^{(i)}(t) \sim \frac{S_l(t)}{\lambda_{a,l}} \quad (3.23)$$

Additionally, insights regarding the previous nomenclature can be derived from Eq. (3.23). For example, as the value of $\lambda_{a,l}$, increases, the update on the l 'th control field decreases. Furthermore, it can be recognised that the strength of the update for each time point is determined by $S_l(t)$.

The shape function and the guess pulse can be used to encode specific features, that are inherited by the optimized pulses. For instance, the boundary condition which was briefly mentioned in section 3.2, would relate to such a feature. In the optimization of quantum gates, shape functions and initial guess pulses can be employed to impose practical constraints, such as a ramping time for a pulse, ensuring that the optimized pulses incorporate this characteristic.

3.4 Gate Implementation

In practice, it is common to determine pulse shapes through OCT methods, with the aim of executing the desired gate in a physical implementation of a quantum computer. This approach often involves formulating quantum gate implementations as optimisation problems. Given a controlled N -qubit array modeled by the Hamiltonian in Eq. (2.14), the goal is to identify a set of control fields $\{u_j(t)\}$ that guide the evolution of the system so that the final

gate $\hat{U}(T, 0, u_j(t))$, generated by the Hamiltonian, closely approximates a desired target gate $\hat{O}$ at the final time $t = T$, with $\hat{O} \in SU(2^N)$. To solve this gate optimization problem, it is convenient to express the objective using functionals. The goal of the optimization is to minimize the functional, which represents the difference between the generated gate and the desired target gate. When the optimization is successfully completed, the functional reaches its minimal value. Consider the functional $J_{T,sm}$, defined as

$$J_{T,sm} = 1 - \left| \sum_{n=1}^N \langle n | \hat{O}^\dagger \hat{U}(T, 0, \epsilon_l) | n \rangle \right|^2. \quad (3.24)$$

This functional captures the optimization objective because, as the generated gate $\hat{U}$ approaches the target gate $\hat{O}$, the functional $J_{T,sm}$ approaches zero. This is due to the fact that $J_{T,sm}$ quantifies the discrepancy between the controlled unitary evolution of the system and the desired target gate $\hat{O}$ ¹. Furthermore, rewriting the sum reveals an intriguing insight, because $J_{T,sm}$ can be expressed in terms of states [51]:

$$1 - \frac{1}{N^2} \left| \sum_{n=1}^N \langle n | \hat{O}^\dagger \hat{U}(T, 0, \epsilon_l) | n \rangle \right|^2 = 1 - \frac{1}{N^2} \left| \sum_{k=1}^N \langle \phi_k^{tgt} | \phi_k(T) \rangle \right|^2.$$

To elaborate, this functional measures the complex overlaps between the target states and the final states at the given control [51]. Consequently, a gate optimization problem can be seen as N simultaneous state optimizations. Additionally, it is important to note that the absolute value is applied outside the sum, which allows for the neglect of the global phase while still enabling control over the relative phases during the optimization process.

In this thesis, gate optimization problems are addressed using numerical implementations of the previously discussed Krotov algorithm. The systems considered are operator-controllable qubit arrays. However, achieving controllability typically requires extended times and ample physical resources. This means that the target gate may not be achievable in arbitrarily short times. Therefore, our focus is on determining the minimum duration required to implement a gate $\hat{O}$ on a controllable qubit array.

$$\tau_{\hat{O}} = \min\{T \geq 0 | \hat{O} \in e^{Lie\{i\hat{H}(t)\}}\}. \quad (3.25)$$

Ultimately, Eq. (3.25) is referred to as the *quantum speed limit* for the unitary gate $\hat{O}$ in a qubit array under a specific configuration of local controls.

To estimate the quantum speed limit of unitary gates, Krotov's method will be employed using a minimal time approach. The optimization task must be clearly defined, with a target gate $\hat{O}$, initial guess pulses $\{u_j(t)\}$, and a specified final time T . Krotov's method is then used to implement $\hat{O}$ at T . The optimization is considered successful when the gate is implemented with the desired level of infidelity, meaning that the value of $J_{T,SM}$ falls below a

¹Although here is a certain functional used, there are various functionals that can be used for gate optimization [50, 51].

certain threshold, such as 10^{-4} . If the optimization is successful, the process is repeated with a shorter final time T_1 . After $N-1$ successful optimizations, it may happen that the algorithm fails to converge in the N th optimization for $t = T_N$, indicating that the desired infidelity cannot be achieved. If this occurs, it might be beneficial to use different initial guess fields and repeat the optimization. Eventually, a candidate for the quantum speed limit will be identified, meaning the desired gate can be implemented with the required infidelity for some $T \geq T_{QSL}$, while for $T < T_{QSL}$, the algorithm fails to converge. For a qubit array subjected to a specific configuration of controls, this approach not only estimates the quantum speed limit of an arbitrary unitary gate but also provides insight into how this quantum speed limit changes when the control configuration is altered.

Chapter 4

Results

In section 4.1, we conduct a series of concise computations on a straightforward two-qubit model. Our objective is to ascertain the maximum speed at which a Bell state can be generated using OCT. Our findings reveal that we can approximate the quantum speed limit with the inverse of the total interaction strengths present in the system. Furthermore, it is anticipated that these findings are connected the speed limits of entangling gates in the forthcoming analysis of 3-qubit systems. The following sections, 4.2, 4.3 and 4.4 present gate speed limits for three distinct qubit architectures. The focus is on chains and loops, as these are the only possible arrangements for three qubits. The aim is to implement a universal gate set comprising Hadamard gates, T gates and B gates, as defined in Eq. (2.34), at the quantum speed limit on the aforementioned chain and loop geometries. The systems under consideration were analysed for single, double, and triple control configuration, as defined in Eqs. (2.29), (2.32) and (2.33). Furthermore, the quantum speed limit was estimated using Krotov's method of optimization in a minimal time approach.

4.1 Preliminary Considerations

Consider two interacting qubits described by the Hamiltonian,

$$\hat{H}_{2q}(t) = \sum_{\alpha=1}^2 -\frac{\omega_{\alpha}}{2} \hat{\sigma}_z^{(\alpha)} + \kappa_a \hat{\sigma}_x^{(1)} \hat{\sigma}_x^{(2)} + u_1(t) \hat{\sigma}_x^{(1)} + u_2(t) \hat{\sigma}_x^{(2)}. \quad (4.1)$$

The qubit frequency and interaction strength are selected to be within a parameter range that is typical for superconducting qubits [8, 12].

$$\frac{\omega_1}{2\pi} = 5.0 \text{ GHz}, \quad \frac{\omega_2}{2\pi} = 5.3 \text{ GHz}, \quad \frac{\kappa_a}{2\pi} = 0.112 \text{ GHz}. \quad (4.2)$$

The objective is to ascertain the shortest duration for the generation of a maximally entangled state. Accordingly, we utilize Krotov's method in a minimal time approach, to estimate the

quantum speed limit of the state transformation:

$$|00\rangle \mapsto |\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle). \quad (4.3)$$

This situation can be compared to the result found by M. Goerz *et al.* in [13]. In the aforementioned research paper, a numerical verification was conducted to demonstrate that the minimum time required to generate a specific maximally entangled state is approximately equal to the inverse of the interaction strength. In this instance, the inverse of the interaction strength is referred to as the *interaction time*. For the Hamiltonian in Eq. (4.1), the interaction time is given by:

$$T_{\kappa_a} = \frac{1}{\kappa_a} \approx 1.421 \text{ ns}. \quad (4.4)$$

Furthermore, Fig. 4.1a illustrates the infidelity as a function of state duration. A sudden change in the state infidelity is observed, which allows us to conclude that the quantum speed limit τ_{QSL} is in the range of 1.1 to 1.5 ns. Moreover, this is consistent with the interaction time, which allows us to postulate that T_{κ_a} provides a reliable estimate for the minimal time required to reach the Bell state when the system is initially in the state of $|00\rangle$. Now, we seek to observe how τ_{QSL} changes, when the coupling in Eq. (4.2) is increased by adding a $\hat{\sigma}_z^{(1)}\hat{\sigma}_z^{(2)}$ - term to Eq. (4.1). Here, Fig. 4.1 shows the OCT results, from which one can see that the minimal time to generate $|\phi^+\rangle$, varies for different coupling types and interaction strengths. In the case of equal interaction strengths, the total qubit-qubit interaction term is given by $\kappa_a(\hat{\sigma}_x\hat{\sigma}_x + \hat{\sigma}_z\hat{\sigma}_z)$, which corresponds to Fig. 4.1b. From there, we can observe that the for the speed limit holds $\tau_{QSL} \in [0.7, 0.8]$ ns.

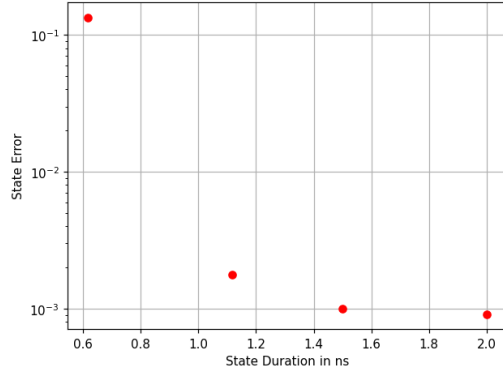
Similar to Eq. (4.4), an attempt is made to estimate the speed limit, by utilising the interaction strengths that are present in the system. Moreover, since two terms constitute to the total qubit-qubit interaction, the inverse of the sum of interaction strengths κ_a is employed to estimate the speed limit:

$$T_{2\kappa_a} = \frac{1}{2\kappa_a} \approx 0.711 \text{ ns}. \quad (4.5)$$

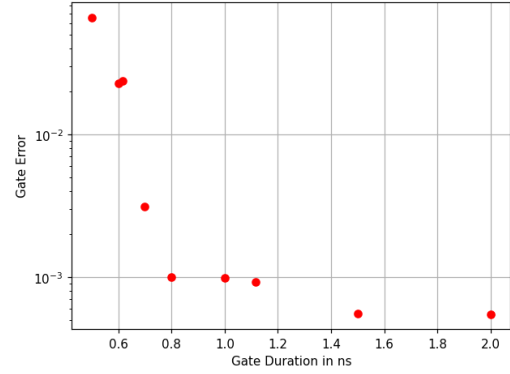
This demonstrates, that the interaction time is consistent with the speed limit indicated in Fig. 4.1b. Furthermore, we consider different prefactors of the qubit-qubit interaction terms. We set $\frac{\kappa_b}{2\pi} = 0.222$ GHz, while $\kappa_a\hat{\sigma}_x\hat{\sigma}_x + \kappa_b\hat{\sigma}_z\hat{\sigma}_z$ describes the total qubit-qubit interaction. As illustrated in Fig. 4.1c, for the speed limit holds, $\tau_{QSL} \in [0.4, 0.5]$. We compare this speed limit to the corresponding interaction time.

$$T_{\kappa_a+\kappa_b} = \frac{1}{\kappa_a + \kappa_b} = \frac{1}{2\pi(0.112 + 0.222)} \approx 0.477 \text{ ns}. \quad (4.6)$$

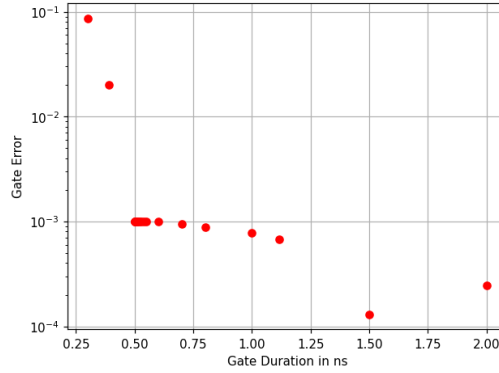
As with the previous instance, the numerical speed limit obtained is consistent with the interaction time. It appears that the inverse of the sum of the interaction strength provides a reliable lower bound for the generation of the Bell state. Furthermore, an examination of all speed limit times presented in Fig. 4.1 demonstrates that increasing the couplings,



(a) $\kappa_a \hat{\sigma}_x \hat{\sigma}_x$ coupling



(b) $\kappa_a (\hat{\sigma}_x \hat{\sigma}_x + \hat{\sigma}_z \hat{\sigma}_z)$ coupling



(c) $\kappa_a \hat{\sigma}_x \hat{\sigma}_x + \kappa_b \hat{\sigma}_z \hat{\sigma}_z$ coupling

Figure 4.1: The infidelity as a function of state duration for $\kappa_a \hat{\sigma}_x \hat{\sigma}_x$ -coupling (a), $\kappa_a (\hat{\sigma}_x \hat{\sigma}_x + \hat{\sigma}_z \hat{\sigma}_z)$ -coupling (b) and $\kappa_a \hat{\sigma}_x \hat{\sigma}_x + \kappa_b \hat{\sigma}_z \hat{\sigma}_z$ -coupling (c).

accelerates the generation of maximal entanglement. To conclude, the minimal time required to generate the Bell state $|\Phi^+\rangle$ can be approximated by the interaction time. It may therefore be surmised that the interaction time represents a lower bound for the implementation of two-qubit gates. Moreover, our findings may indicate that adding interactions may also influence the rate at which information is transferred between qubits.

4.2 Asymmetric Chain

The asymmetric chain is a straightforward system for modelling superconducting qubits. For this reason, the qubit frequencies and the qubit-qubit interaction strengths were chosen in a typical parameter regime. Thus, the asymmetric chain has the qubit frequencies

$$\frac{\omega_1}{2\pi} = 5.0 \text{ GHz}, \quad \frac{\omega_2}{2\pi} = 5.3 \text{ GHz}, \quad \frac{\omega_3}{2\pi} = 5.5 \text{ GHz}. \quad (4.7)$$

and the interaction strengths

$$\frac{\kappa_{(x,x)}^{(1,2)}}{2\pi} \approx 0.112 \text{ GHz}, \quad \frac{\kappa_{(z,z)}^{(1,2)}}{2\pi} \approx 0.222 \text{ GHz}, \quad \frac{\kappa_{(x,x)}^{(2,3)}}{2\pi} \approx 0.314 \text{ GHz}. \quad (4.8)$$

With regard to the asymmetric chain, it is not immediately evident which characteristics determine the rate at which information propagates through the system or a gate is executed. It seems reasonable to propose that a consequence of non-uniform interactions is the emergence of stronger and weaker parts of the chain. Accordingly, information may propagate rapidly through sections exhibiting strong interactions, but may then subsequently decelerate significantly upon reaching a section with a weaker interaction. In such a scenario, the weaker side is likely to act as a bottleneck, limiting the transfer of information and prolonging the implementation of gates.

The left side exhibits a greater number of couplings than the right side of the chain. This could lead to the assumption that the less coupled right side must be the weaker side and consequently acting as a bottleneck. A further argument for the proposition that the right side must be the weaker side, is provided by an overall consideration of the total interaction strengths on each side.

$$\kappa_{left} = \kappa^{(1,2)} = \kappa_{(x,x)}^{(1,2)} + \kappa_{(z,z)}^{(1,2)} \approx 2\pi \cdot 0.334 \text{ GHz} > 2\pi \cdot 0.314 \text{ GHz} = \kappa_{(x,x)}^{(2,3)} = \kappa_{right}. \quad (4.9)$$

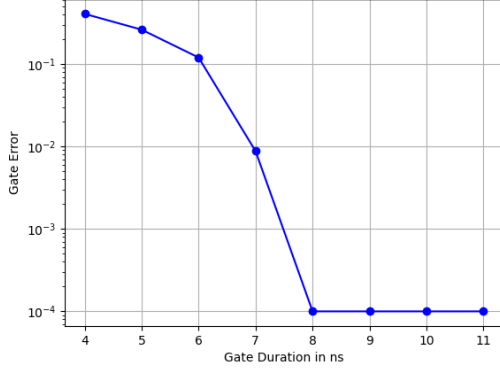
Although the considered universal set comprises nine gates, only a few gate speed limits may be relevant in estimating the runtime of a quantum protocol. Given that the asymmetric chain is universal, any quantum algorithm expressed through a sequence of universal gates, such as that described in Eq.(2.34), is executable on that system. Although the order of gates in an arbitrary quantum algorithm is unknown, it is nevertheless possible to estimate its runtime by performing a worst-case estimation, that is, by considering the highest of all gate durations. In this context, high gate speed limits are of particular significance when estimating the minimal time required to execute an arbitrary quantum algorithm on a universal qubit array. It follows that the maximum of all gate speed limits is defined as the quantum speed limit of the universal set $\mathcal{U}$. Formally, one can write

$$\tau_{max} = \max\{\tau_{U_1}, \tau_{U_2}, \dots, \tau_{U_M}\}, \quad U_i \in \mathcal{U}. \quad (4.10)$$

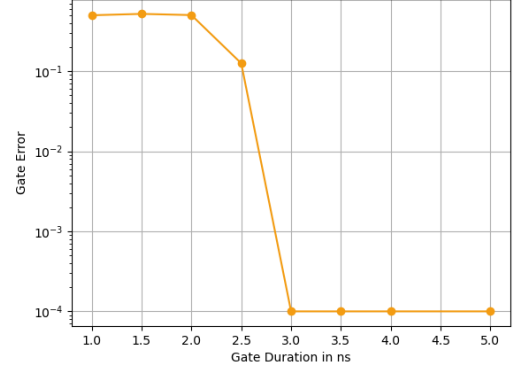
Furthermore, it would be intriguing to investigate the extent to which the quantum speed limit of the universal set τ_{max} , differs from the other gate speed limits. Accordingly, we define

	CP1	CP2	CP3	CP12	CP13	CP23	CP123
$\bar{\tau}$ [ns]	7.1	4.53	8.3	2.2	2.05	2.32	0.6
τ_{max} [ns]	9	8	10	3	3.5	4	2
U_{max}	$B^{(2,3)}$	$H^{(3)}$	$T^{(2)}, T^{(3)}$	$B^{(1,3)}, H^{(3)}$	$H^{(2)}$	$B^{(1,3)}$	$B^{(1,3)}$

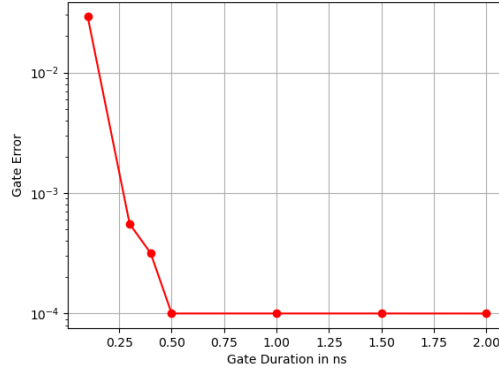
Table 4.1: Results for asymm. chain.



(a) CP2, $\tau_{H^{(3)}} = 8$ ns.



(b) CP12, $\tau_{H^{(3)}} = 3$ ns.



(c) CP123, $\tau_{H^{(3)}} = 0.3$ ns.

Figure 4.2: Illustrated is the infidelity $J_{T,sm}$ as a function of gate duration for the implementation of $H^{(3)}$ using CP2 (a), CP12 (b) and CP123 (c). Furthermore, this comparison shows that the quantum speed limit reduces, when the number of controls increases.

$\bar{\tau}$ as the average value of the speed limits computed for all gates from the universal set $\mathcal{U}$.

$$\bar{\tau} = \frac{1}{M} \sum_{i=1}^M \tau_{U_i}. \quad (4.11)$$

Table 4.1 illustrates τ_{max} and $\bar{\tau}$, for every control configuration. Note that τ_{max} and $\bar{\tau}$ decrease in the number of local controls. Furthermore, U_{max} represents the gates with the longest duration, which dictate the speed limit τ_{max} . Additionally, it is evident that various quantum gates dictate the quantum speed limit. Nevertheless, it is possible that a gate dictates τ_{max} for multiple configurations. For instance, $H^{(3)}$ sets τ_{max} for CP2 and CP12. To continue with the example of the $H^{(3)}$ gate implementation, a comparison of its quantum speed limit in single, double and triple control setups is presented in Fig. 4.2. The gate infidelity as a function of gate duration is shown, for the CP2 setup in Fig. 4.2a. This curve

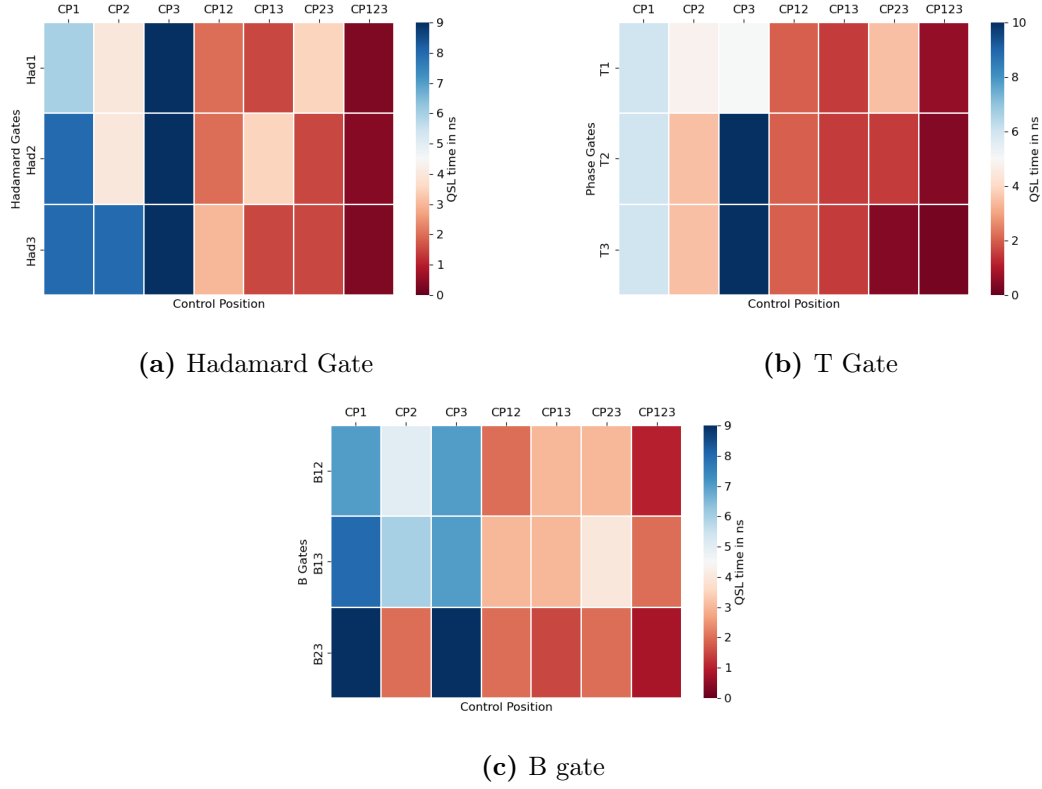


Figure 4.3: For the asymmetric chain, the heatmaps display the quantum speed limits for each gate within the universal set as defined in Eq. (2.34).

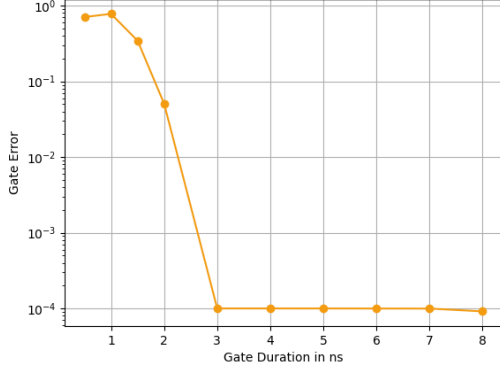
provides the gate speed limit $\tau_{H^{(3)}} = 8$ ns, since for durations below, implementations with desired infidelity $J_{T,sm} < 10^{-4}$ are not reached. By comparing the infidelities provided in Fig. 4.2, one concludes that the speed limit candidate for the CP12 configuration is between those for CP2 and CP123. The gate speed limits are collectively presented in Fig. 4.3, which provides a comprehensive overview that allows to view and compare the speed limits of each gate. Each heat map pertains to a gate type (e.g., Hadamard) and assigns colors to all local gates of this type (vertical axis) based on the control configuration (horizontal axis), representing the corresponding speed limits. The color spectrum ranges from dark blue (indicating relatively long gate times) to dark red (indicating relatively short gate times). In the following, we are going to discuss the quantum speed limits in Table 4.1 and elaborate on physical reasons that could explain the long gate durations of U_{max} , for all control configurations. With regard to CP1, it has been observed that the implementation of gates targeting the first qubit is more rapid, which suggests that direct control of the target may prove advantageous. Furthermore, the speed limit in this configuration is dictated by the $B^{(2,3)}$ gate. The qubits that are the primary focus of this gate implementation are not directly controlled and are situated on the right-hand side of the asymmetric chain. It thus follows that the pulse must propagate from the first qubit through the left-side coupling

before reaching the centre, where it will activate the second qubit. Subsequently, the pulse propagates through the right-side coupling, ultimately reaching the right end of the chain. Therefore, the time required for the pulse to propagate through the system must elapse which contributes to the gate duration. Furthermore, the implementation of that gate represents a significant challenge for the local control. Because, it must realize the desired entanglement between the qubits (2) and (3), while it must decouple the controlled qubit (1) from the remaining qubits. In considering the constraints of our resources, it is reasonable to anticipate that the minimum time required for this process will be considerably longer than the fastest implementation of that gate, which occurs in triple control configuration at 0.8 ns. The speed limit for the CP2 configuration is set by $H^{(3)}$. The gate implementation represents another instance of indirect control of the target qubit, since it requires a Hadamard gate implementation on the right end of the chain. Hence, pulse propagation times contribute to the gate duration. One could argue further, that the right side is slightly weaker (in terms of total coupling strength), which implies even longer pulse propagation times and hence longer gate durations. However, this can not be the entire reason, since other gates that target the third qubit do not demonstrate prolonged gate times. We find that

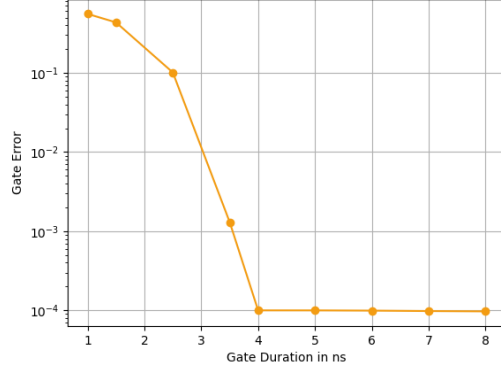
$$\tau_{T^{(3)}} = 3.5 \text{ ns} < \tau_{T^{(1)}} = 4.8 \text{ ns}.$$

To identify the importance of the coupling terms for this implementation process, we consider the implementation of $H^{(1)}$ (in the same configuration), which is analogous to the $H^{(3)}$ implementation a case of indirect target control. Although the total interaction strength on the left side $\kappa^{1,2}$ is only slightly stronger, it holds that $\tau_{H^{(1)}} = 4 \text{ ns}$. Therefore, we conjecture that the $\hat{\sigma}_z \hat{\sigma}_z$ -interaction is of importance for the Hadamard gate implementation. Furthermore, it can be proposed that the absence of a $\hat{\sigma}_z^{(2)} \hat{\sigma}_z^{(3)}$ interaction, causes the different gate durations for $H^{(1)}$ and $H^{(3)}$. This example will be revisited in the following section, where the implications of another $\hat{\sigma}_z^{(2)} \hat{\sigma}_z^{(3)}$ -interaction will be discussed in greater detail, see section 4.3. The case of CP3 appears to be more complicated. It is unexpected that gates targeting the third qubit exhibit prolonged gate times and impose a limiting speed with $T^{(3)}$. This is surprising given that directly controlling the target was previously considered a key component for fast implementation. It could be the case that the implementation of the gate is slowed down if the local control is applied to the qubit that is less coupled to the remaining system. Ultimately, the challenge of the CP3 is to implement the T gate on the third qubit while counteracting the system's intrinsic entanglement. Furthermore, the rapid entangling evolution of qubits (1) and (2), due to their strong coupling, necessitates greater resources to disentangle them. The extended gate duration of $T^{(2)}$ might result from the interplay between the indirect gate implementation and suboptimal positioning of the local control on the less coupled qubit.

Moreover, it is noteworthy that the gate duration of $T^{(1)}$ is the shortest at 5 ns, despite the target qubit being the furthest away. This is particularly surprising given that the other T gates have the longest gate durations and set the speed limit in this setup. However, the speed up could stem from the target qubit being at the more coupled side of the chain. In regard to the speed limits in single control configuration, it can be concluded that all gates that



(a) In CP12 setup, $\tau_{B^{(1,3)}} = 3$ ns.



(b) In CP23 setup, $\tau_{B^{(1,3)}} = 4$ ns.

Figure 4.4: Infidelity curves of $B^{(1,3)}$ implementation in certain double control configurations. The CP12 speed limit and the CP23 speed limit are dictated by the $B^{(1,3)}$ gate, since it exhibits the longest gate durations for both control setups.

dictate the speed limit are either targeting the third qubit or, in the case of the $T^{(2)}$ gate, are indirectly controlled through the single $\hat{\sigma}_x \hat{\sigma}_x$ -coupling with the third qubit. It can be inferred that the less coupled right side of the chain is often responsible for longer gate durations. For the same reason, the average speed limit for gates, implemented with the CP3 configuration is the most time expensive among any other single control configuration. Furthermore, this can be observed by considering the CP3 column in the heat maps 4.3. Moreover, this provides justification for referring to CP3 as a cold control. Conversely, CP2 is regarded as a hot control due to the shortest average gate times among all single control setups. In view of the quantum speed limit in double control configuration, it can be concluded that the prolonged gate durations of $B^{(1,3)}$ and $H^{(3)}$ in the CP12 configuration are to be expected. Indeed, this case may be analogous to the speed limit observed in the CP2 configurations. As both gate implementations target the qubit on the less coupled side of the chain, pulse propagation times contribute to the gate duration. Regarding the CP13 setup, the $H^{(2)}$ gate dictates τ_{max} . The implementation task is an example of indirect control, which necessitates the propagation of pulses from both chain ends to the center. Moreover, the distinct couplings and strengths of the interactions on either side result in the propagation of pulses at varying rates. As a result, when the pulses from the right and left sides arrive at the centre, they can interact in a way that incurs significant costs. Regarding the prolonged gate time observed in the CP23 configuration is the indirect implementation of the target on the remaining qubit through the control of the less coupled side.

It is important to note that the speed limits associated with the double control configuration relate to instances of indirect gate implementations, which is also a common occurrence in the single control configuration. Nevertheless, it appears that the role of the entangling gates is becoming more pronounced when additional controls are employed, as the speed limit in CP12 and CP23 configuration is determined by the $B^{(1,3)}$ gate. The underlying

reasons for this observation will be discussed subsequently in this section. Here, Fig. 4.4a to 4.4b illustrates the infidelity as function of gate duration, for the aforementioned double control configurations. Furthermore, the example presented in Fig. 4.4 also illustrates that altering the positions of both local controls impacts the gate speed limit, likely due to variations in $\hat{\sigma}_x\hat{\sigma}_x$ and $\hat{\sigma}_z\hat{\sigma}_z$ interactions. In the triple control configuration any gate quantum speed limit the shortest and consequently its implementation is the fastest. Furthermore, τ_{max} is dictated by $B^{(1,3)}$. To provide a detailed explanation of this observation, we examine the implementation of all entangling gates at the quantum speed limit. Our analysis reveals notable similarities to the minimal time to prepare a Bell state in Eq. (4.3). For the $B^{(1,2)}$ gate holds that

$$T_{\kappa^{(1,2)}} = 0.476 \text{ ns} < T_{\kappa_{zz}^{(1,2)}} = 0.717 \text{ ns} < \tau_{B^{(1,2)}} = 1 \text{ ns} < T_{\kappa_{xx}^{(1,2)}} = 1.421 \text{ ns}. \quad (4.12)$$

Furthermore, for the $B^{(2,3)}$ gate holds that

$$T_{\kappa_{zz}^{(2,3)}} = 0.509 \text{ ns} < \tau_{B^{(2,3)}} = 0.8 \text{ ns}. \quad (4.13)$$

Hence, the interaction times provide lower bounds for the quantum speed limits of the B gates. It is noteworthy that the $B^{(1,2)}$ speed limit can be more precisely estimated by employing the interaction times of the individual coupling terms, rather than the inverse of the total interaction strength. It is notable that, a connection has been established between the B gate speed limits and the interaction strengths present in the system. A comparison of the B gate speed limits in triple control configuration demonstrates the following:

$$\tau_{B^{(1,2)}} + \tau_{B^{(2,3)}} = 1.8 \text{ ns} < 2 \text{ ns} = \tau_{B^{(1,3)}} = \tau_{max}. \quad (4.14)$$

It is not stated that the implementation process of the $B^{(1,3)}$ gate necessitates the prior implementation of the other B gates. Nevertheless, the duration of $B^{(1,2)}$ and $B^{(2,3)}$ provide a timescale for the execution of an entangling operation for qubits (1) and (2) and qubits (2) and (3). The gate duration of $B^{(1,3)}$, is the longest among all gates, due to the fact that its implementation necessitates time-consuming sequential steps, which themselves require resources for execution. A protocol for implementing the B gate could be as follows: first, entangle qubits (1) and (2), then entangle qubits (2) and (3), and finally disentangle qubit (2) from the others, leaving qubits (1) and (3) entangled. In conclusion, the $B^{(1,3)}$ speed limit can be related to the $B^{(1,2)}$ and $B^{(2,3)}$ speed limits, which themselves can be related to interactions strength that are present in the system.

Furthermore, it can be posited that the high average gate duration for CP3 and CP23 and the prolonged gate durations of $H^{(3)}$ in the CP2 and CP12 setup, are associated with the less coupled right side of the asymmetric chain. Hence, the less coupled side functions as a bottleneck, extending the time required to implement certain gates. Therefore, a speed limit analysis on another chain that features a $\hat{\sigma}_z^{(2)}\hat{\sigma}_z^{(3)}$ coupling is performed, to verify the accuracy of the hypothesis.

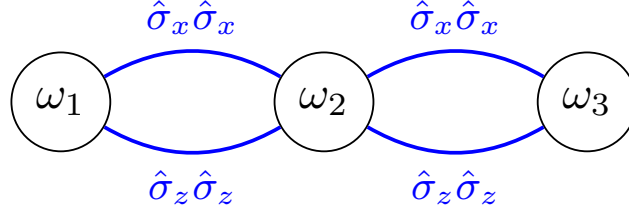


Figure 4.5: Symmetric chain

4.3 Symmetric Chain

Increasing the coupling between the second and the third qubits by adding a $\hat{\sigma}_z\hat{\sigma}_z$ -interaction transforms the previously discussed qubit array into the symmetric chain. Qubit frequencies and interaction strengths are chosen according to the Eqs. (4.7) and (4.8). The Hamiltonian describing the symmetric chain reads

$$\hat{H}_S = \hat{H}_A + \kappa_{(z,z)}^{2,3} \hat{\sigma}_z^{(2)} \hat{\sigma}_z^{(3)}, \quad \frac{\kappa_{(z,z)}^{2,3}}{2\pi} = 0.156 \text{ GHz}. \quad (4.15)$$

The investigation of the symmetric chain allows to study the effect of the $\hat{\sigma}_z^2\hat{\sigma}_z^3$ - coupling term on the quantum speed limits. Given that the LHS and the RHS of the symmetric chain exhibit the same number and type of couplings, it will be the aim of this analysis to identify the bottleneck in this system. Furthermore, it may provide insight into whether the absence of $\hat{\sigma}_z^2\hat{\sigma}_z^3$ - coupling term resulted in a delay for specific gate durations.

A comparison of the speed limits listed in Table 4.2 and Table 4.1 reveals several noteworthy observations. First, τ_{max} decreases for CP2, CP3 and all double control configurations. With regard to $\bar{\tau}$, it is evident that alterations are apparent in each configuration, whether substantial or trivial. Furthermore, it can be observed that the vast majority of speed limits displayed in Table Fig. 4.2 are determined by the $B^{(1,3)}$ speed limit, with the exception of the CP3 speed limit, which is determined by $H^{(3)}$. These points will be discussed in greater detail subsequently. The most significant alterations to single controls are observed in CP1 and CP3. This is further corroborated by an examination of the heat maps in Fig. 4.6. To elaborate further, an average improvement in CP3 is observed, while CP1 shows an average deterioration. The additional interaction seems to slow down CP1. Moreover, CP3 shows faster implementation in the symmetric chain, whereas it performed the worst

	CP1	CP2	CP3	CP12	CP13	CP23	CP123
$\bar{\tau}$	8.1	4.72	5.7	2.05	2.01	2.31	0.7
τ_{max}	9	6	6.5	2.5	3.5	3.5	2
U_{max}	$B^{(1,3)}, B^{(2,3)}$	$T^{(1)}, B^{(1,2)}, B^{(1,3)}$	$H^{(1)}$	$B^{(1,3)}$	$B^{(1,3)}$	$B^{(1,3)}, T^{(1)}$	$B^{(1,3)}$

Table 4.2: Results for symm. chain.

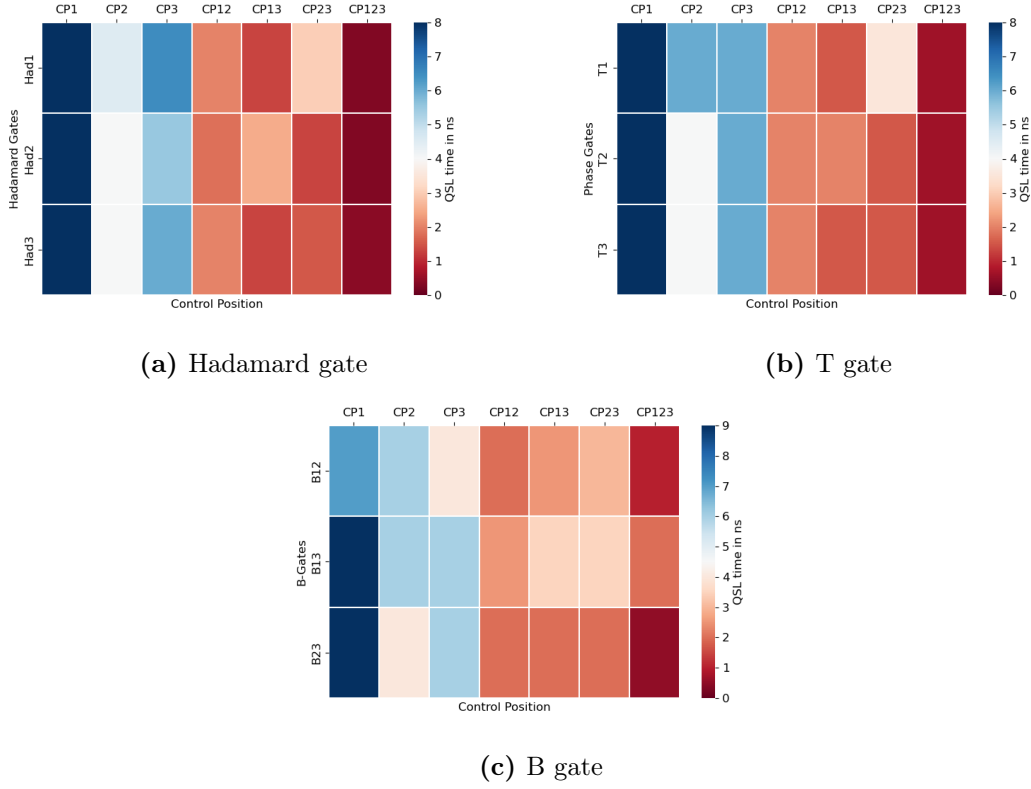
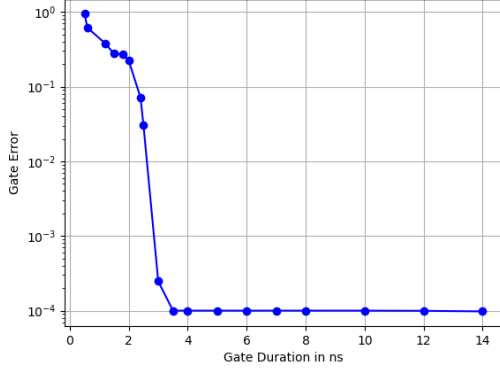


Figure 4.6: For the symmetric chain, the heatmaps illustrate the quantum speed limits for each gate within the previously mentioned universal set.

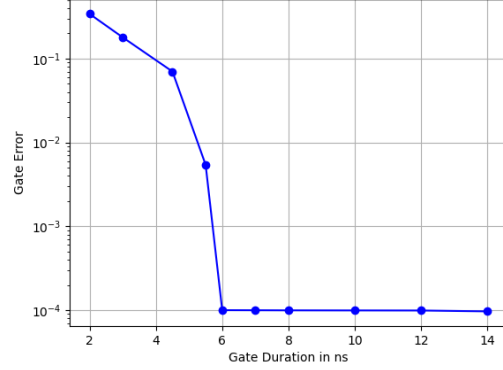
in the asymmetric chain. This prompts the question of what is responsible for the observed improvement in CP3 and deterioration in CP1, given that they appear to exhibit opposing qualities. This result may be explained by analyzing the total interaction strength on each side.

$$\kappa^{1,2} = \kappa_{xx}^{1,2} + \kappa_{zz}^{1,2} = 2\pi \cdot 0.334 < 2\pi \cdot 0.470 = \kappa_{xx}^{2,3} + \kappa_{zz}^{2,3} = \kappa^{2,3}. \quad (4.16)$$

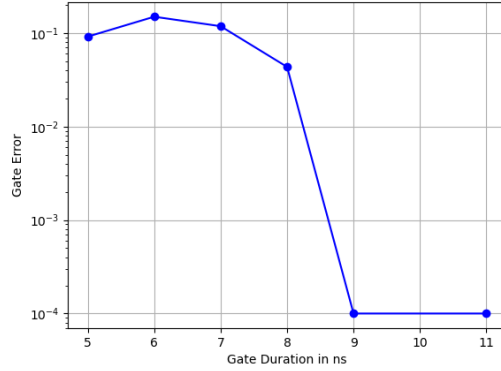
Therefore, the integration of the aforementioned coupling into the system results in a relocation of the maximum total coupling energy from the left to the right side of the symm. chain. This could indicate that the transfer of information between qubits 3 and 2 occurs at a faster rate than that between qubits 1 and 2. It may therefore be posited that CP3 represents an ineffective control position, given that the third qubit is connected to the system via the weakest couplings. It can be concluded that the additional coupling shifts the maximum coupling energy to the right side, thereby enhancing the performance of CP3. In the following we investigate some speed limits for the aforementioned configurations and elaborate on potential reasons for deteriorations and improvements in comparison to the asymmetric chain. Regarding CP1, we note that $B^{(2,3)}$ dictates the CP1 speed limit with the same value as in the asymmetric chain. Hence, the additional coupling between qubits (2) and (3), is not accelerating the implementation of that gate. Furthermore, it should be noted that



(a) $\tau_{H^{(3)}} \approx 4$ ns .



(b) $\tau_{H^{(3)}} = 6$ ns .



(c) $\tau_{H^{(3)}} = 9$ ns .

Figure 4.7: Infidelity curves of $H^{(3)}$ implementation for symm. chain in CP2 (a), and CP3 (b). Furthermore, for the asymm. chain in CP3 (c).

both, the asymm. chain and the symm. chain, have identical left sides. This leads to the assumption that the physical mechanism that leads to this speed limit is presumably more impacted by the left side of the symmetric chain. Thus, the explanation concerning the gate duration is likely to be analogous to that of the symmetric chain. It is hypothesised that the limiting factor in this gate implementation is the process of disentangling qubit (1) from the remaining qubits. A more detailed explanation of why this could be the case is given in the discussion. Moreover, the speed limit is also determined by $B^{(1,3)}$, which has demonstrated a decline in comparison to the previous qubit array.

To further elaborate, a reduction in τ_{max} for CP2 and CP3 is observed, as well as a change in U_{max} . For instance, $H^{(3)}$ does not dictate τ_{max} for the CP2 configuration (see Fig. 4.2a). In the asymmetric chain this gate exhibited a speed limit of 8 ns. In contrast, $H^{(3)}$ demonstrates a significantly reduced gate speed limit of 4 ns in the symmetric chain as illustrated in Fig. 4.7a. One can conclude, that the additional interaction accelerates the information transfer

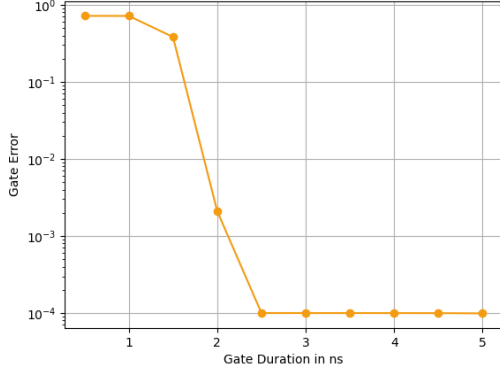
to the third qubit, which enhances the implementation process. The gates $B^{(1,2)}$ and $T^{(1)}$ show a deterioration regarding their duration in comparison to the asymmetric chain. Furthermore, they determine the speed limit in the CP2 configuration. This can be attributed to indirect control and the fact that targeting the side with less total coupling strength in the symmetric chain is relatively time-consuming. It is surprising, however, that the gate durations increase despite the left side of the symmetric chain being identical to the left section of the asymmetric chain. Additionally, it might seem unexpected that an extra coupling between qubits (2) and (3) slows down the entanglement of qubits (1) and (2). Nevertheless, this extra coupling also speeds up the entanglement between qubits (2) and (3), which might require additional resources for disentangling.

In double control setups, we see that $H^{(3)}$ does not determine τ_{max} for CP12, as the additional coupling, similar to the CP2 configuration, facilitates faster information transfer. On the other hand, $B^{(1,3)}$ involves indirect gate implementation and, due to the required steps for its generation, is expected to have longer gate durations. Additionally, the $B^{(1,3)}$ gate sets the speed limit in the CP13 configuration, where the duration actually decreases compared to the asymmetric chain.

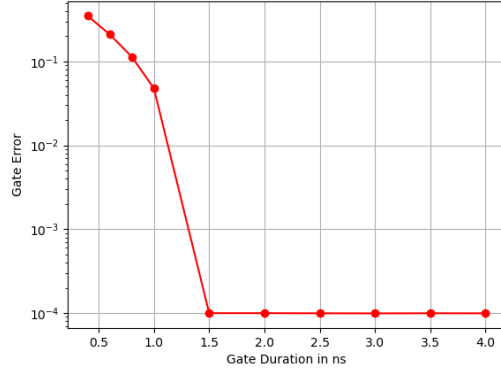
By examining Table 5.2, one can observe a minor difference of just half a nanosecond between the speed limits of the CP12 and CP123 configurations. Both of these speed limits are determined by the $B^{(1,3)}$ gate, as illustrated by the infidelity curves in Fig. 4.8a and Fig. 5.1b. This suggests that, despite having additional resources in the form of local controls in the triple control configuration, their efficiency is only marginally improved. Therefore, we hypothesize that inherent system parameters significantly hinder the generation of the B gate. We aim to understand why the $B^{(1,3)}$ gate also sets τ_{max} in the symmetric chain for the triple control configuration. To address this, we analyze the remaining entangling gates in the triple control setup, similar to our approach in the previous section. We hypothesize that it is not surprising that the speed limit of $B^{(1,2)}$ does not decrease, as the bottleneck in this case appears to be related more to the coupling strength on the left side, as indicated in Eq. (4.12). Therefore, the implementation of the $B^{(1,2)}$ gate may be nearly identical to that in the asymmetric chain. Additionally, we observe a slight improvement in the speed limit of the $B^{(2,3)}$ gate, approximately 0.3 ns. This enhancement is anticipated, given our conjecture about the relationship between interaction time and the quantum speed limit of the B gate, as described in Eq. (4.13). The additional coupling increases the total interaction strength between qubits (2) and (3), thereby reducing the interaction time. Since it provides a lower bound for $\tau_{B^{(2,3)}}$ gate, it holds that

$$T_{\kappa^{2,3}} = 0.339 \text{ ns} < 0.5 \text{ ns} = \tau_{B^{(2,3)}}. \quad (4.17)$$

Since the additional interaction speeds up the entanglement generation between qubits (2) and (3), one might expect it to also accelerate the corresponding sequential step in the $B^{(1,3)}$ gate implementation. However, no improvement in the $B^{(1,3)}$ speed limit is observed. Additionally, we note that $\tau_{B^{(1,3)}}$ decreases for CP3, CP12, and CP23, but increases for CP1 and CP13. This suggests that potential deteriorations and improvements could be balanced by utilizing all available controls during the implementation process. Given the expectation



(a) CP12 setup, $\tau_{B^{(1,3)}} = 2.5$ ns.



(b) CP123 setup, $\tau_{B^{(1,3)}} = 1.5$ ns.

Figure 4.8: Infidelity curve of $B^{(1,3)}$ implementation, for CP12 (a) and CP123 (b).

that additional couplings would accelerate the formation of entanglement in the analysed system, we reached the unexpected conclusion that this is not always the case. In certain instances, the execution of entangling gates can be completed more rapidly. However, in other instances, the duration of the gate or the occurrence of changes is reduced. Despite the expectation that additional couplings would speed up entanglement formation in the analyzed system, we reached the unexpected conclusion that this is not always true. In some cases, entangling gates can indeed be executed more quickly, while in other cases, the duration of the gate or the extent of changes is reduced. Notably, the $B^{(1,3)}$ gate often imposed a limit on the speed of the universal set in most instances. To conjecture why the $B^{(1,3)}$ gate dictates the speed limits for nearly every control configuration, we suspect that the newly integrated coupling strength, $\kappa_{zz}^{(2,3)}$, complicates the entangling phases. This additional coupling may require the system more time to naturally evolve towards an entangling gate closer to the B gate. Consequently, the control mechanisms must fine-tune the process to achieve the desired implementation, affecting the overall speed limit. This raises the question of whether improving the performance of the $B^{(1,3)}$ gate by incorporating an interaction that couples the first and third qubit could be a viable approach.

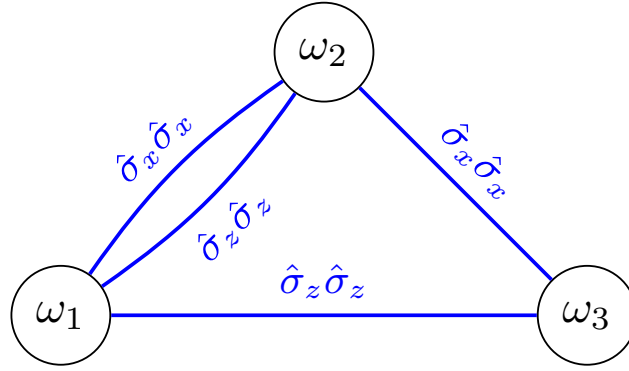


Figure 4.9: Qubit Loop

4.4 Qubit Loop

Creating a coupling between the first and the third qubits via an $\hat{\sigma}_z \hat{\sigma}_z$ -interaction transforms the asymmetric chain into a qubit loop described by the Hamiltonian

$$\hat{H}_L = \hat{H}_A + \kappa_{(z,z)}^{1,3} \hat{\sigma}_z^{(1)} \hat{\sigma}_z^{(3)}, \quad \kappa_{(z,z)}^{1,3} = 0.156 \text{ GHz}. \quad (4.18)$$

This is the first system under consideration that exhibits all-to-all connectivity, whereby each qubit is connected to all the remaining qubits. This could give rise to the emergence of intriguing intrinsic physical properties, different from those observed in qubit chains. In a chain configuration, a local operation on a qubit at one end has a direct impact on the subsequent neighbour. However, as the pulse propagates through the chain, its effects gradually extend to the other end. In a qubit loop, there are no ends, and thus the implementation of a local operation on any qubit will have indirect implications for all other qubits. To illustrate, if one controls qubit (3), then one directly affects qubit (2) via the operator $\kappa_{xx}^{2,3} \hat{\sigma}_x \hat{\sigma}_x$ and qubit (1) via the interaction represented by the operator $\kappa_{zz}^{1,3} \hat{\sigma}_z \hat{\sigma}_z$. Furthermore, since qubits (1) and (2) are coupled they might influence each other further, which could result in more complicated dynamics.

However, regarding the total coupling strength it holds that

$$\kappa^{1,2} \approx 2.14 \kappa^{1,3}, \quad \kappa^{2,3} \approx 2.01 \kappa^{1,3}. \quad (4.19)$$

Therefore, the coupling strength between qubits (1) and (3) is the weakest. Additionally, the corresponding interaction times are analyzed to gain insights into how this coupling affects

	CP1	CP2	CP3	CP12	CP13	CP23	CP123
$\bar{\tau}$	9.1	6.6	8.2	2.12	1.82	2.38	0.83
τ_{max}	11	10	10	3	3	4	1.5
U_{max}	$H^{(3)}$	$T^{(3)}$	$T^{(2)}, T^{(3)}$	$H^{(3)}, B^{(1,3)}$	$H^{(2)}$	$B^{(1,3)}$	$B^{(1,3)}$

Table 4.3: Results for qubit loop.

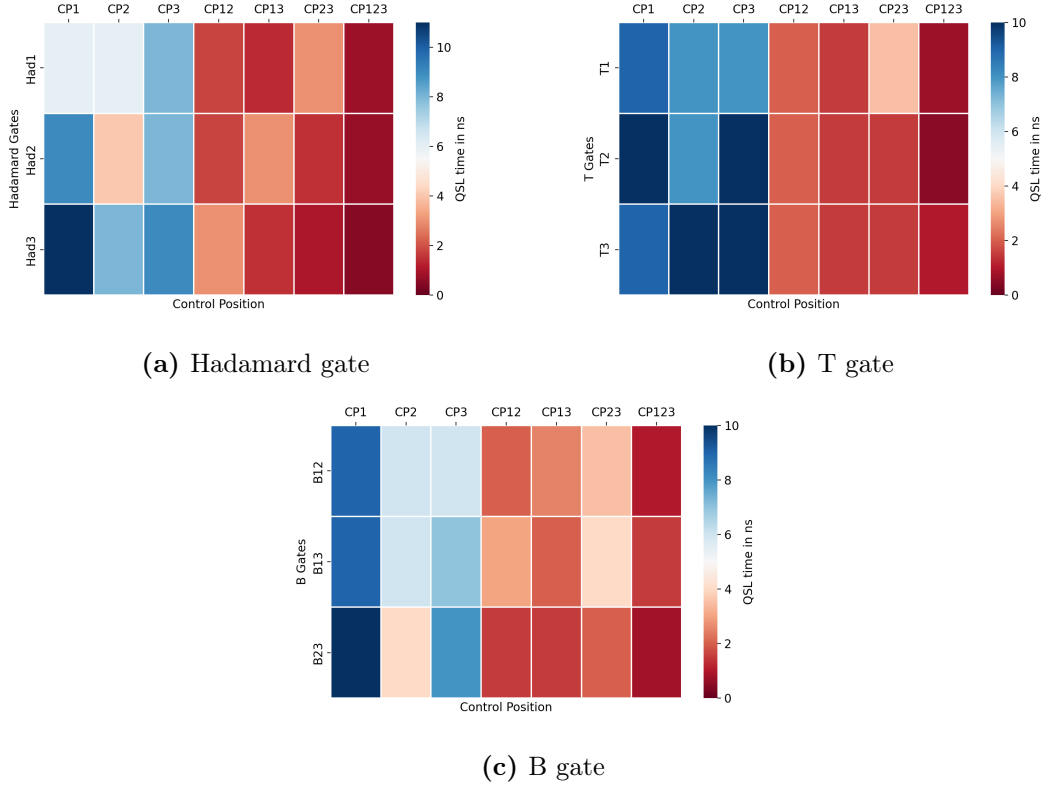


Figure 4.10: For the qubit loop, the heatmaps depict the quantum speed limits for each gate in the specified universal set.

entanglement and information transfer rates. There, it holds that

$$T_{\kappa^{1,3}} \approx 1.02 \text{ ns} > 0.989 \text{ ns} = T_{\kappa^{1,2}} + T_{\kappa^{2,3}}. \quad (4.20)$$

It can be inferred that the relatively low strength of the loop coupling leads to a correspondingly low information transfer rate. It would be interesting to determine whether the mere presence of the coupling is significant, regardless of its strength, and whether there are situations where the coupling can be practically ignored. In the latter scenario, the loop would behave similarly to the asymmetric chain, as the coupling between qubits (1) and (3) is so weak that it might be negligible in some cases. The following section will compare the data from the loop with that from the qubit chains, discussing any discrepancies and their potential explanations. Compared to the asymmetric chain, there is a noticeable degradation in the speed limits for CP1 and CP2. In the qubit loop, the speed limit for CP1 is determined by $H^{(3)}$, and it decreases by about 3 nanoseconds compared to both chains. This loop interaction facilitates the transfer of information between qubits (1) and (3). It can be concluded that, due to the weak interaction, the propagation rate is slow, necessitating a slower gate implementation relative to the chains.

In the CP2 configuration, the $T^{(3)}$ gate sets τ_{max} , and it is over six nanoseconds slower com-

pared to the chains. This delay arises because the pulse must propagate from the controlled qubit to the remaining qubits, traversing two separate pathways to reach the third qubit. The more time-consuming and costly pathway is likely responsible for this delay. It is reasonable to hypothesize that the pathway with the greater number of nodes—moving from qubit (2) to (1) to (3)—contributes to the delay. Additionally, the low propagation rate between qubits (1) and (3) further impacts the overall gate time. It is also noteworthy that the average gate time in this configuration is decreasing, which supports the idea that this reasoning may apply to other gates as well.

It should be noted that for the remaining control configurations, U_{max} corresponds to the same set of gates as in the asymmetric chain. This supports the hypothesis that the weak loop interaction does not significantly alter the system and that the loop behaves similarly to the asymmetric chain in many instances. However, there are slight increases in speed of about half a nanosecond for the CP13 and CP123 configurations. In the case of CP13, it was suggested that the delayed arrival of pulses at qubit (2) might be challenging for the control. Nevertheless, because the loop interaction couples the controlled qubits (1) and (3), it could enhance control stability and therefore facilitate gate implementation. Moreover, concerning the speed limit of the universal set in the triple control configuration, the loop interaction allows for a more straightforward entanglement of qubits (1) and (3) without the need for time-consuming additional steps, unlike in the chains. Additionally, the loop interaction time serves as a close lower bound for the quantum speed limit of $B^{(1,3)}$ in this configuration.

$$T_{\kappa^{1,3}} \approx 1.02 \text{ ns} < 1.5 \text{ ns} = \tau_{B^{(1,3)}}. \quad (4.21)$$

This may be taken as an indication that our hypothesis could prove to be correct.

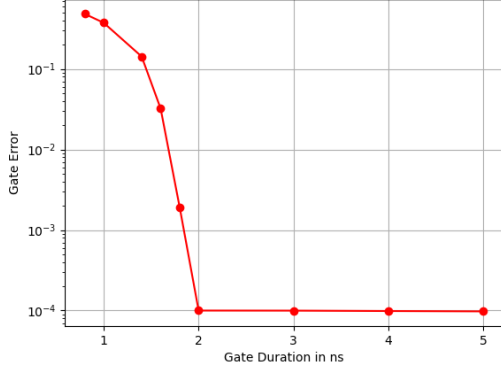
Chapter 5

Discussion and Outlook

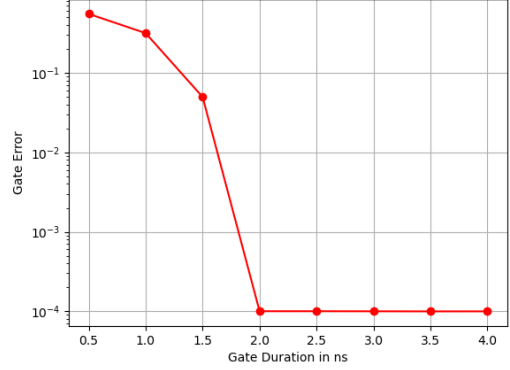
In this master thesis, the quantum speed limit of a universal set consisting of H, T and B was estimated, for two 3-qubit chains and one 3-qubit loop. Each system was analyzed under seven distinct control configurations, and the search for the speed limit was performed using Krotov's method with a minimal time approach. In regard to the gate times, these have been found to range between 11 and 0.2 ns across all geometries. The quantum speed limit of any unitary gate is the shortest and consequently its implementation is the quickest, in the case of every qubit being locally controlled (maximal local controls are utilized). The results show that the quantum speed limit of the universal set, denoted by τ_{max} , varies with the number of utilized local controls and where they are placed in the system. Our findings also show that, internal system properties set a lower bound on τ_{max} , which showcases that the highest rate at which (external) local controls steer the system is dictated by internal parameters, which becomes more evident, the more local controls are utilized. In the case of maximal local controls (CP123), the $B^{(1,3)}$ gate dictates the speed limit of the universal set τ_{max} for all considered qubit architectures, see Fig. 5.1. Moreover, the speed limit of that gate takes the same value for both 3-qubit chains:

$$\tau_{B^{(1,3)}} = \tau_{max}^{ASYM} = \tau_{max}^{SYM} = 2 \text{ ns.} \quad (5.1)$$

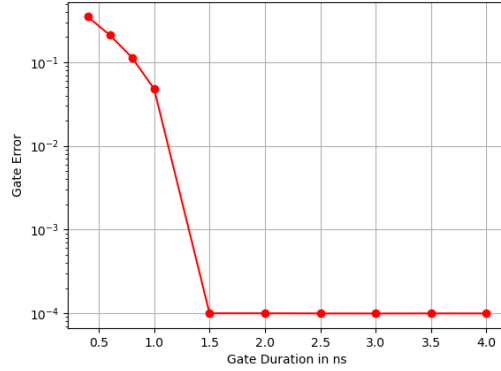
We propose that the total time to implement the $B^{(1,3)}$ gate is influenced by the duration of the three steps required for its execution, as discussed in Chapter 4.2. For both the asymmetric and symmetric chains, it was shown that $\tau_{B^{(1,2)}} + \tau_{B^{(2,3)}} < \tau_{B^{(1,3)}}$, which suggests that implementing $B^{(1,3)}$ may necessitate prior entanglement between qubits (1) and (2), as well as between qubits (2) and (3). The execution time of $B^{(1,3)}$ is 0.5 nanoseconds faster on the qubit loop. This improvement may be due to the presence of $\hat{\sigma}_z^{(1)}\hat{\sigma}_z^{(3)}$, which could accelerate entanglement generation between qubits (1) and (3) by reducing the number of sequential steps required, compared to the chains. Furthermore, this speed limit can also be lower-bounded by the corresponding interaction time, as indicated in Eq. (4.21). Therefore, we conclude that internal parameters constrain the maximum speed of the B gate and any entangling operations in the system. To clarify the relationship between internal parameters, such as interaction strengths, and gate quantum speed limits, it is helpful to consider the



(a) Asymm. Chain, $\tau_{B^{(1,3)}} = 2$ ns.



(b) Symm. Chain, $\tau_{B^{(1,3)}} = 2$ ns.



(c) Loop, $\tau_{B^{(1,3)}} = 1.5$ ns.

Figure 5.1: The infidelity as a function of gate duration is shown for the asymmetric chain (a), the symmetric chain (b), and the qubit loop (c). It has become evident that the $B^{(1,3)}$ gate sets the speed limit for the universal set $\mathcal{U}$ across all the qubit architectures considered for max. local controls.

interaction strengths in the system as a frequency of entangling evolution. For instance, the quantum speed limit of the $B^{(2,3)}$ gate in the CP1 configuration of both the asymmetric and symmetric chains was estimated to be 9 ns. We suspect that the left side of the chain, which is identical for both chains, dictates the physical mechanism responsible for this quantum speed limit. Therefore, we consider only the dynamics induced by the couplings:

$$\hat{U}(t) = e^{-i\kappa_{xx}^{(12)} \hat{\sigma}_x \otimes \hat{\sigma}_x t} e^{-i\kappa_{zz}^{(12)} \hat{\sigma}_z \otimes \hat{\sigma}_z t}. \quad (5.2)$$

Furthermore, we note that at $T = \frac{2\pi}{\kappa_{xx}^{(12)}} \approx 8.93$ ns, it holds that $\hat{U}(T) = 1$. Thus, the entanglement between qubits (1) and (2) naturally disappears at this time. Based on this and the previous reasoning, we can conjecture that CP1 steers the entanglement between qubits (2) and (3) while waiting for the system to naturally decouple qubits (1) and (2). This

could explain the prolonged gate time of the $B^{(2,3)}$ gate. In conclusion, this approach may provide greater insight into the physical nature of the obtained gate speed limits.

Furthermore, regarding the emergence of long gate durations, several factors contribute to time delays. Often, gates that determine the quantum speed limit target qubits that are not directly controlled. However, targeting an uncontrolled qubit does not necessarily guarantee a long gate time. Various intrinsic system properties can affect gate durations. For example, in the asymmetric chain, the unequal distribution of interaction terms results in a more coupled side and a less coupled side. It has become evident that the less coupled side acts as a bottleneck during gate implementation. In the symmetric chain, we frequently observed performance deterioration with entangling gates rather than faster entanglement generation. We suspected that the coupling strength $\kappa_{zz}^{(1,3)}$ to complicate the entangling phases. Regarding the qubit loop, the weak loop interaction created a delay in gate duration, particularly for implementing local gates with only a single control.

Moreover, the values of gate speed limits could be subjected to more rigorous mathematical analysis by examining the Lie algebra elements generated at depth d . Several sources suggest that the time required to execute a gate is directly proportional to the depth of its generator [35]. Lie algebra elements at depth d were examined for the asymmetric chain under various control configurations. For the CP1 configuration, verification using Optimal Control Theory (OCT) confirmed that the speed limit is determined by the $B^{(2,3)}$ gate. Operators of the form $\hat{\sigma}_x^{(2)}\hat{\sigma}_x^{(3)}$ and were found admissible at high depth. A similar analysis for the CP2 configuration showed that the speed limit for gate $H^{(3)}$ could be generated by an operator of the form:

$$G = -i\frac{\pi}{2}\frac{1}{\sqrt{2}}(\hat{\sigma}_x^{(3)} + \hat{\sigma}_z^{(3)}). \quad (5.3)$$

The analysis revealed that all Pauli operators acting on the third qubit were only fully admissible at high depths, with σ_x^3 being the only element present at the highest depth. However, the presence of suspected generators at high depth does not alone justify a long execution time. In the CP3 case, local operators of the form $\hat{\sigma}_y^3$ and $\hat{\sigma}_z^3$ are accessible at depth 2, yet the $T^{(3)}$ gate dictates the speed limit for the universal set, demonstrating the limitations of this argument. Despite early emergence of its generator, the number and structure of early depth Lie algebra elements might be inadequate for controlling the system, potentially leaving insufficient control directions. Therefore, considering generator depths as a future approach may be valuable, provided the mathematical arguments are more thoroughly validated.

Regarding the potential connection between fast gate implementation and the maximum Lie algebra depth, a comparison of the speed limits in Table 4.1 with the highest Lie algebra depth, as shown in Fig. 2.1, reveals that the highest depth does not clearly predict which configuration will achieve the highest speed limit. However, it is observed that the Lie algebra depth decreases as the number of local controls increases.

Looking ahead, further testing of the obtained speed limits is necessary. Combining gradient-free pre-optimization with Krotov's method for fine-tuning has been shown to yield better results than using either method in isolation [14]. Additionally, testing speed limit candidates against random pulses could be beneficial. This approach aims to approximate a global

search through the extensive use of multiple pulses [16].

The next step is to scale up the system size and conduct a speed limit search for larger qubit arrays. While extensive data has been collected for various 5-qubit configurations, it was not included in this thesis due to scope constraints. From the analysis of 3-qubit geometries with maximal controls, it was observed that the $B^{(1,3)}$ gate sets the speed limit. This suggests that for longer qubit chains with maximal local controls, the speed limit will likely be determined by the minimum time required to maximally entangle the qubits that are the furthest apart. The next milestone involves making the model more realistic by incorporating tunable couplers in qubit arrays. Many quantum processors feature time-dependent, tunable qubit-qubit interactions [17], which can enhance implementation speed. For instance, when implementing a local gate on a specific qubit, couplings between other qubits can be turned off to avoid unintended entanglement. Ideally, to entangle two qubits, only their coupling is active, while others are deactivated. However, the feasibility of achieving a universal set speed limit with such tunable interactions remains uncertain, as it often requires either infinite resources and extended times. The challenge is determining how long it will take to generate the full dynamical Lie algebra or the gates of the universal set when couplings are adjusted. This issue has not been addressed in current literature. The project's next step is to explore the speed limit of the universal set while accounting for the dynamics of open systems. Including effects like dephasing and amplitude damping introduces non-unitary evolution, raising questions about whether these factors will slow down gate implementation or increase the speed limit, and how these effects compare to the unitary case. One motivation for this project was to balance the number of local controls with implementation speed. The symmetric chain in the CP12 configuration has a speed limit that is nearly identical to the CP123 configuration, differing by only half a nanosecond. It would be valuable to explore whether this system could be used as a foundational element for scaling up the system. Additionally, examining how the quantum speed limit changes with system size would be of great interest.

The clock speed of a quantum computer is defined as the maximum rate at which quantum protocols can be executed on its processor. This thesis aimed to understand the theoretical minimum time for gate implementation by analyzing gate speed limits. However, achieving gates at their quantum speed limit is only one aspect of the overall clock speed. Other factors also impact the clock speed, such as qubit initialization, measurement, readout, and reset processes. Even if gates are executed at their quantum speed limit, the overall clock speed may not significantly improve if these other processes are comparatively lengthy. In an ideal scenario, where all processes occur at their theoretical minimum times, the cycle time would reach its quantum speed limit, maximizing the clock speed. It would be valuable to explore how the clock speed of systems like the symmetric chain varies with different control configurations. Additionally, using OCT to determine the minimal times for initialization, measurement, readout, and reset can help estimate the shortest possible cycle time based on the number of local controls.

Appendix

These tables list the quantum speed limits for each gate across all considered control configurations, measured in nanoseconds.

Table 5.1: Asymmetric Chain

	CP1	CP2	CP3	CP12	CP13	CP23	CP123
$H^{(1)}$	6.0	4.0	9.0	2.0	1.5	3.5	0.3
$H^{(2)}$	8.0	4.0	9.0	2.0	3.5	1.5	0.4
$H^{(3)}$	8.0	8.0	9.0	3.0	1.5	1.5	0.3
$T^{(1)}$	6.0	4.8	5.0	2.0	1.5	3.5	0.6
$T^{(2)}$	6.0	3.5	10.0	2.0	1.5	1.5	0.4
$T^{(3)}$	6.0	3.5	10.0	2.0	1.5	0.4	0.2
$B^{(1,2)}$	7.0	5.0	7.0	2.0	3.0	3.0	1.0
$B^{(1,3)}$	8.0	6.0	7.0	3.0	3.0	4.0	2.0
$B^{(2,3)}$	9.0	2.0	9.0	2.0	1.5	2.0	0.8

Table 5.2: Symmetric Chain

	CP1	CP2	CP3	CP12	CP13	CP23	CP123
$H^{(1)}$	8.0	4.5	6.5	2.0	1.3	3.0	0.3
$H^{(2)}$	8.0	4.0	5.5	1.8	2.5	1.3	0.3
$H^{(3)}$	8.0	4.0	6.0	2.0	1.3	1.5	0.4
$T^{(1)}$	8.0	6.0	6.0	2.0	1.5	3.5	0.6
$T^{(2)}$	8.0	4.0	6.0	2.0	2.0	1.5	0.6
$T^{(3)}$	8.0	4.0	6.0	2.0	1.5	1.5	0.6
$B^{(1,2)}$	7.0	6.0	4.0	2.0	2.5	3.0	1.0
$B^{(1,3)}$	9.0	6.0	6.0	2.5	3.5	3.5	2.0
$B^{(2,3)}$	9.0	4.0	6.0	2.0	2.0	2.0	0.5

Table 5.3: Qubit Loop

	CP1	CP2	CP3	CP12	CP13	CP23	CP123
$H^{(1)}$	6.0	6.0	8.0	1.8	1.4	3.0	0.8
$H^{(2)}$	9.0	4.0	8.0	1.8	3.0	1.5	0.7
$H^{(3)}$	11.0	8.0	9.0	3.0	1.5	1.0	0.5
$T^{(1)}$	9.0	8.0	8.0	2.0	1.5	3.5	0.7
$T^{(2)}$	10.0	8.0	10.0	2.0	1.5	1.5	0.5
$T^{(3)}$	9.0	10.0	10.0	2.0	1.5	1.5	1.0
$B^{(1,2)}$	9.0	6.0	6.0	2.0	2.5	3.5	1.0
$B^{(1,3)}$	9.0	6.0	7.0	3.0	2.0	4.0	1.5
$B^{(2,3)}$	10.0	4.0	8.0	1.5	1.5	2.0	0.8

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